

MATH 27300 Lecture Notes

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May 4, 2026

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1 Tuesday, September 30

1.1 Background

Definition 1.1.1 (Differential equation). A differential equation takes the form $F[x, y, \partial_i y, \partial_i \partial_j y, \dots, \partial^{(n)} y]$.

Some examples are the heat equation ($\partial_t u = \sum_{i=1}^n \partial_i^2 u$, a PDE) and $\frac{d^2}{dt^2} u + \frac{d}{dt} u = u$, an ODE.

Definition 1.1.2 (Ordinary, partial differential equation). An ODE contains one derivative, while a PDE contains at least 2 different partial derivatives.

Definition 1.1.3 (Order). The order of an ODE is the number of its highest-order derivative.

Definition 1.1.4 ((Non)linear). An ODE is linear if $F[x, y, y'(x), \dots, y^{(n)}(x)]$ depends on $y, y', \dots, y^{(n)}$ linearly. It is nonlinear otherwise. The ODE is linear even if it depends on x nonlinearly. A linear ODE is in the form $a_n(x)y^{(n)}(x) + \dots + a_0(x)y = b(x)$.

1.2 Linear Differential Equations

Consider the linear first-order ODE in the form

$$\frac{d}{dt}y = C.$$

C is a constant; we have the solution of a family $y = Ct + B$ for $B \in \mathbb{R}$.

If we are given $\frac{d^2}{dt^2}y = 0$, our solution is $y = Ct + B$ for $C, B \in \mathbb{R}$. These are called general solutions.

The solutions are not unique equations. If we impose an initial condition in the first-order case of $y(t_0) = y_0$ with a given t_0, y_0 , then the solution is unique. In the second-order case, we generally need two initial conditions of $y(t_0) = y_0$ and $y'(t_0) = y'_0$.

Consider the general first-order linear ODE $a_1(t)y'(t) + a_0(t)y(t) = b(t)$.

(First case)

$$\begin{aligned} \frac{d}{dt}[y(t)] &= g(t) \\ y(t) &= \int g(t)dt. \end{aligned}$$

(Second case) If we drop the $b(t)$ term and want to solve $\frac{d}{dt}y(t) =$

$a(t)y(t)$, we separate into unknowns and gives:

$$\begin{aligned}\frac{\frac{d}{dt}y(t)}{y(t)} &= a(t) \\ \frac{d}{dt} \underbrace{[\ln |y|]}_{z(t)} &= a(t), \quad \text{if } y(t) \neq 0 \\ \implies \ln [|y|] &= \int a(t)dt + C \\ |y| &= \exp \left[\int a(t)dt \right] \cdot e^C \\ y &= K \cdot \exp \left[\int a(t)dt \right],\end{aligned}$$

where $K \in \mathbb{R}$ ($K = 0$ covers the case when $y(t) = 0$).

(General case) In the general case, we cannot just divide by $y(t)$ because $\frac{b(t)}{y(t)}$ remains. But for $\mu(t) \neq 0$,

Try finding $a(t)$ such that $\frac{d}{dt} [a(t)y(t)] = \mu y'(t) + \mu p(t)y(t)$.

$$\mu(t)y'(t) + \mu(t)p(t)y(t) = \mu(t)g(t)$$

tells us $\frac{d}{dt}\mu(t) = \mu(t)p(t)$ by the product rule, so

$$\mu(t) = \exp \left[\int_{t_0}^t p(s)ds \right] \quad (\text{Second case})$$

$$\text{Hence, } \frac{d}{dt} \left[\exp \left[\int_{t_0}^t p(s)ds \right] y(t) \right] = \frac{d}{dt} [\mu(t)y(t)] = \mu(t)g(t).$$

We have $\frac{d}{dt} [\mu y] = \mu g$ and are in the first case, so

$$\begin{aligned}\mu(t)y(t) &= \int_{t_0}^t \mu(s)g(s)ds + C \\ y(t) &= \frac{1}{\mu(t)} \int_{t_0}^t \mu(s)g(s)ds + C.\end{aligned}$$

Example 1.2.1. Suppose we want to solve $y' + y = e^t$. We want to find a suitable μ for $\mu(t)y' + \mu(t)y = \mu(t)e^t$.

This resembles the result of the product rule, so we need $\frac{d}{dt} [\mu y] = \mu y' + \mu' y$, so $\mu' = \mu$, $\mu(t) = e^t$. So

$$\begin{aligned}\frac{d}{dt} [\mu y] &= \frac{d}{dt} [e^t y] = e^{2t} \\ y &= \frac{1}{2}e^t + C \cdot e^{-t}.\end{aligned}$$

To determine C , we put $y(t_0) = y_0$ from an initial condition and know $\mu y(t) = \int_{t_0}^t \mu g(s)ds + C$. Taking $t = t_0$ and putting $\mu(t) = \exp \left[\int_{t_0}^t g(s)ds \right]$ to make integration easier,

$$\begin{aligned}\mu y(t_0) &= C \\ \mu(t_0) = 1 &\implies C = y(t_0) = 1.\end{aligned}$$

Hence,

$$y(t) = \frac{1}{\mu(t)} \left[\int_{t_0}^t \mu(s)g(s)ds + y_0 \right].$$

2 Thursday, October 2

2.1 Review

Recall that for a first-order linear ODE $y'(t) + p(t)y = g(t)$, we can write $F[x, y, y'(x)] = 0$. By the Implicit Function Theorem, we can write $y'(x) = f(x, y)$.

2.2 Separation of Variables

Put $N = 1$ and $m = -f$, so we have

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0.$$

If we can write $M = M(x)$ and $N = N(y)$, then call the ODE separable, so we can formally separate into

$$N(y)dy = -M(x)dx$$

and integrate if N and M have primitives. We end with $n(y) = m(x) + C$ and can solve for y .

Rigorously,

$$\begin{aligned} M(x) + N(y) \frac{dy}{dx} &= 0 \\ \int M(x)dx + \int [N(y) \frac{dy}{dx}]dx &= 0 \\ \int M(x)dx + \int N(y)dy &= 0. \end{aligned}$$

Example 2.2.1. We want to solve $x + y \frac{dy}{dx} = 0$. Then $M = x, N = y$.

$$\begin{aligned} \frac{x^2}{2} + C + \int y \frac{dy}{dx} dx &= 0 \\ \frac{x^2}{2} + C + \frac{y^2}{2} &= 0. \end{aligned}$$

From this quadratic, we will get two solutions, where $y(x) = y_1(x), y_2(x)$. With an initial condition, we get one solution.

We can manipulate the starting ODE to make it separable.

Example 2.2.2. Consider $y + e^x \frac{dy}{dx} = 0$. We want M to be a function of just x , so divide everything by ye^x .

$$\begin{aligned} e^{-x} + \frac{1}{y} \frac{dy}{dx} &= 0 \\ -e^{-x} + C + \ln |y| &= 0. \end{aligned}$$

More generally, suppose we have $M_1(x)M_2(y) + N_1(x)N_2(y)\frac{dy}{dx} = 0$. We can divide by M_2N_1 to get

$$\frac{M_1}{N_1}(x) + \frac{N_2}{M_2}(y)\frac{dy}{dx} = 0.$$

Example 2.2.3. Consider

$$e^{x+y} + xy\frac{dy}{dx} = 0.$$

Then divide by $e^y x$ to get

$$\frac{e^x}{x} + \frac{y}{e^y}\frac{dy}{dx} = 0.$$

This seems to be a solution method for a large class of functions, but consider when we have a larger sum

$$M_1(x)M_2(y) + Z_1(x)Z_2(y) + N_1(x)N_2(y)\frac{dy}{dx} = 0.$$

Let's return to trying to manipulate our ODE into the form

$$\frac{d}{dx} \left[\underbrace{\phi(x, y(x))}_{\text{unknown}} \right] = \underbrace{g(x)}_{\text{known}}.$$

Then

$$\begin{aligned} \frac{d}{dx} \left[\phi(x, y(x)) - \int g(x) dx \right] &= 0 \\ \frac{d}{dx} \tilde{\phi}(x, y(x)) &= 0. \end{aligned}$$

WLOG, take $g = 0$.

$$\begin{aligned} \frac{d}{dx} \phi(x, y(x)) &= 0 \\ \partial_1 \phi + \partial_2 \phi \frac{dy}{dx} &= 0 \\ \implies M(x, y) = \partial_1 \phi(x, y), N &= \partial_2 \phi(x, y). \end{aligned}$$

This motivates the following because...

Definition 2.2.4 (Exact). An ODE is exact if it is in the form $M(x, y) + N(x, y)\frac{dy}{dx} = 0$ and there exists a function ϕ such that $M(x, y) = \partial_1 \phi(x, y), N(x, y) = \partial_2 \phi(x, y)$.

For the ODE to be exact, $\partial_2 M = \partial_1 N = \partial_1 \partial_2 \phi$ must hold.

Theorem 2.2.5. Suppose $M, N, \partial_2 M, \partial_1 N$ are continuous in $B = [a, b] \times [c, d]$. The ODE $M(x, y) + N(x, y)\frac{dy}{dx} = 0$ is exact if and only if

$$\partial_2 M = \partial_1 N,$$

i.e. if we can find a function ϕ such that $\partial_1 \phi = M$ and $N = \partial_2 \phi$.

This is supposed to be the most general functional form we want to solve?

Remark 2.2.6. Separable ODEs are exact, so this theorem is more powerful than our previous solution methods.

If we have a separable ODE, $\partial_2 M(x, y) = 0 = \partial_1 N(x, y)$, so it is exact.

Proof. **necessity?** For sufficiently, suppose that $\partial_2 M(x, y) = \partial_1 N(x, y)$. We want to construct ϕ such that $\partial_1 \phi = M$ and $\partial_2 \phi = N$.

If ϕ satisfies $\partial_1 \phi = M$, the following holds for any h such that $\partial_1 h(x, y) = 0$:

$\partial_1 \phi(x, y) = M(x, y)$, and integrating over x

$\phi(x, y) = \int M(x, y) dx + h(y)$ since we only integrated over x

$$\phi(x, y) = \int_{x_0}^x M(s, y) ds + h(y).$$

Now we want $\partial_2 \phi = N$.

$$\begin{aligned} \partial_2 \phi(x, y) &= \int_{x_0}^x \partial_2 M(s, y) ds + h'(y) \\ &= \int_{x_0}^x \partial_1 N(s, y) ds + h'(y) \\ &= N(x, y) - N(x_0, y) + h'(y). \end{aligned}$$

We get $\partial_2 \phi = N$ if we can find h such that $h'(y) = N(x_0, y)$.

$$h(y) = \int_{y_0}^y N(x_0, t) dt$$

works.

So,

$$\phi(x, y) = \int_{x_0}^x M(s, y) ds + \int_{y_0}^y N(x_0, t) dt.$$

□

Example 2.2.7. Consider $y^2 x + (1 + x^2 y) \frac{dy}{dx} = 0$. Put $M(x, y) = y^2 x$ and $N(x, y) = 1 + x^2 y$. Then $\partial_2 M = 2xy = \partial_1 N$, so this ODE is exact.

Following the theorem's steps, we have

$$\phi(x, y) = \int y^2 x dx = \frac{x^2 y^2}{2} + h(y).$$

We want $\partial_2 \phi = N$, so

$$\begin{aligned} x^2 y + h'(y) &= 1 + x^2 y \\ h'(y) &= 1 \implies h(y) \\ \phi(x, y) &= \frac{x^2 y^2}{2} + y, \frac{d}{dx} \phi(x, y(x)) = 0. \end{aligned}$$

Hence $\frac{x^2 y^2}{2} + y = C$ and we have two solutions $y_1(x; C), y_2(x; C)$ and use an initial condition to pick one, either plugging the IC into the solution (more natural) or solving $y(x_0) = y_0$ for C .

where the does the last equation come from

The following will not be tested.

Multiplying by a function μ may result in $\mu M + \mu N \frac{dy}{dx} = 0$ is exact ($\partial_2(\mu M) = \partial_1(\mu N)$), even if $M + N \frac{dy}{dx} = 0$ is not. If μ depends on both x, y then we end up with PDEs, so consider μ only depending on x or y .

In the former case,

$$\begin{aligned}\mu \partial_2 M &= \frac{d}{dx} \mu(x) N + \mu \partial_1 N \\ N \frac{d}{dx} \mu(x) &= \mu [\partial_2 M - \partial_1 N] \\ \underbrace{\frac{\frac{d}{dx} \mu(x)}{\mu(x)}}_{\text{unknown}} &= \underbrace{\frac{\partial_2 M - \partial_1 N}{N}}_{\text{known}}.\end{aligned}$$

The LHS is a function of x and the RHS is generally a function of x, y . This is solvable iff the RHS is a function of x only.

3 Tuesday, October 7: Second-Order Linear ODEs

3.1 General Second-Order Linear ODEs

If we have a second-order ODE $F(t, y, y', y'') = 0$ and F depends on its arguments linearly, then we can write

$$P(t)y'' + Q(t)y' + R(t)y = G(t) \quad (1)$$

$$y'' + p(t)y' + q(t)y = g(t), \quad (2)$$

where the second form is more common (so long as $P(t) \neq 0$). The initial conditions take on the form $y(t_0) = y_0, y'(t_0) = y'_0$.

If $y'' = 0$, then our solution is linear:
 $y(t) = c_1 t + c_2$.

We call the ODE homogeneous if $g = 0$. This is because if y solves the ODE, then ay also solves it for any $a \in \mathbb{C}$. The ODE is nonhomogeneous if $g \neq 0$.

3.2 The simplest case

For now, assume we have a second-order constant-coefficient homogeneous ODE in the form

$$ay'' + by' + cy = 0$$

for $a, b, c \in \mathbb{R}$.

(Case 1: $a = 0$) Our equation becomes $by' + cy = 0$, so $y = Ce^{-\frac{c}{b}t}$.

(Case 2: $a \neq 0$) We try an exponential function, since taking a derivative leads to a very similar function. Starting with the guess $y(t) = c_0 e^{\lambda(t)}$, we plug in y, y', y'' to get

$$c_0 [a\lambda^2 e^{\lambda t} + b\lambda e^{\lambda t} + ce^{\lambda t}] = 0. \quad (3)$$

Dividing by c_0 , if λ is a root of $a\lambda^2 + b\lambda + c = 0$, then y is a solution of the ODE.

We get two roots to the quadratic, λ_1 and λ_2 , which lead to two solutions for the ODE: $y_1(t) = c_1 e^{\lambda_1 t}$, $y_2(t) = c_2 e^{\lambda_2 t}$.

Denote the differential operator $L[\cdot]$ where

$$L[y] = P(t)y'' + Q(t)y' + R(t)y = 0.$$

We can show that for constants c_1, c_2 and functions y_1, y_2 ,

$$L[c_1 y_1 + c_2 y_2] = c_1 L[y_1] + c_2 L[y_2]$$

with a short computation.

Remark 3.2.1. (Principle of Linear Superposition) Note that y solves equation 2 iff $Ly = 0$. Hence, if both y_1 and y_2 solve equation 2, then for any constants c_1, c_2 , $y(t) = c_1 y_1(t) + c_2 y_2(t)$ also solves equation 2.

We call y is the general solution because it incorporates both specific solutions y_1, y_2 .

We call the specific solutions y_1, y_2 "different" if they are linearly independent: $y_2(t) \neq c y_1(t)$.

Example 3.2.2. Consider $y'' - 5y' + 6y = 0$. Writing $\lambda^2 - 5\lambda + 6 = 0$, we get the two roots of $\lambda_1 = 3$ and $\lambda_2 = 2$. Then our solutions are different:

$$\begin{aligned} y_1(t) &= e^{2t} \\ y_2(t) &= e^{3t}. \end{aligned}$$

Our general solution is

$$y(t) = c_1 e^{2t} + c_2 e^{3t}.$$

We are interested in getting two *real* solutions to 3, since this implies that we can scale by any real c_1, c_2 and obtain a real-valued solution. However, this is not the only possibility:

- There are two different real roots $\lambda_1, \lambda_2 \in \mathbb{R}$.
- $\lambda_2 = \overline{\lambda_1}$. In this case, the solution $e^{\lambda_1 t}$ is not real.
- $\lambda_1 = \lambda_2$. The previous case implies that λ must be real.

3.3 Complex Roots of the Characteristic Equation

Ideally, we could transform the second case so that $y(t)$ is actually real by choosing c_1, c_2 intelligently. Suppose we have a solution $y(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$ with complex λ_1, λ_2 . Then

$$\begin{aligned} e^{\lambda_1 t} &= e^{(\lambda + i\mu)t} = e^{\lambda t} e^{i\mu t} \\ &= e^{\lambda t} [\cos(\mu t) + i \sin(\mu t)] \\ e^{\lambda_2 t} &= e^{\lambda t} [\cos(\mu t) - i \sin(\mu t)]. \end{aligned}$$

Note that the solutions are different iff the quadratic in equation 3 has two unique roots.

How do we choose c_1, c_2 such that $y(t) \in \mathbb{R}$? Choosing $c_1 = c_2$, we get

$$y_1(t) = 2c_1 e^{\lambda t} \cos(\mu t).$$

The easy choice is $c_1 = \frac{1}{2}$, making $y_1(t) = e^{\lambda t} \cos(\mu t)$

If we pick $c_1 = -c_2$, then we get

$$y_2(t) = 2ic_1 e^{\lambda t} \sin(\mu t).$$

Then the easy choice is $c_1 = \frac{1}{2i}$, making $y_2(t) = e^{\lambda t} \sin(\mu t)$

Putting these together, we get the general solution

$$y(t) = (\tilde{c}_1 \cos(\mu t) + \tilde{c}_2 \sin(\mu t)) e^{\lambda t},$$

where any $c_1, c_2 \in \mathbb{R}$ keep $y(t) \in \mathbb{R}$!

Example 3.3.1. Consider $y'' + y = 0$. Then solving $\lambda^2 + 1 = 0$, we get $\lambda_1 = i, \lambda_2 = -i$. Hence $\lambda = 0, \mu = 1$. Then the general solution is

$$y(t) = c_1 y_1(t) + c_2 y_2(t) = c_1 \cos(t) + c_2 \sin(t).$$

3.4 Repeated Roots; Reduction of Order

review this section in the book.

Now consider the third case, where we have a unique real solution (that is repeated in the quadratic). We know in $a\lambda^2 + b\lambda + c = 0$, $4ac = b^2$, so $\lambda_1 = \lambda_2 = -\frac{b}{2a}$. $y_1(t) = e^{\lambda_1 t}$, and now we want to find another solution. To do so, we will apply a reduction of order – going from an n th order ODE to an $n - 1$ th order ODE.

We conjecture the form $y_2(t) = \mu(t)y_1(t)$ because μ' solves the first order ODE.??

Suppose y_1 solves

$$p(t)y_1'' + q(t)y_1' + r(t)y_1 = 0.$$

Expanding, we see that

$$0 = P(\mu y_1)'' + Q(\mu y_1)' + R(\mu y_1)$$

$$0 = P[\cdot] + Q[\dot{\cdot}] + R[\dot{\cdot}]$$

and combining, the μ -term gives $\mu [Py_1'' + Qy_1' + Ry] = 0$.

Hence, we get something like $A(t)\mu'' + B(t)\mu' = 0$, which is actually a first-order linear ODE for $z(t) := \mu'(t)$.

$\mu'' p y_1 + \mu' [2P y_1' + Q y_1] = 0$ can be solved for $\mu'(t)$, then further integrated to recover μ .

Example 3.4.1. Suppose $P = a, Q = b, R = c, y_1 = e^{\lambda_1 t}, \lambda_1 = -\frac{b}{2a}$.

Then $y_2(t) = (c_1 t + c_2) e^{\lambda_1 t} = t e^{\lambda_1 t}$. So the general solution is

$$y(t) = (c_1 t + c_2) e^{\lambda_1 t}.$$

4 Thursday, October 9: Series Solutions of Second-Order Linear ODEs

4.1 Series Solutions Near an Ordinary Point

Last time, we assumed that $ay'' + by' + cy = 0$ had constants a, b, c , so $y = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$. Now, we will replace these constants with functions P, Q, R of x and find solutions to

$$P(x)y'' + Q(x)y' + R(x)y = 0 \quad (4)$$

in the form

$$y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n. \quad (5)$$

We will introduce the notion of convergence, where $\mu := \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}$ dictates that if $|x - x_0| < \mu^{-1}$, equation (5) converges.

Convergence is important because where y is analytic,¹

¹ $x \in B(x_0, \mu^{-1})$

$$y'(x) = \sum_{n \geq 1} a_n n (x - x_0)^{n-1}$$

and

$$y^{(k)}(x) = \sum_{n \geq k} a_n n \cdots (n - k + 1) (x - x_0)^{n-k}.$$

We will want to plug in these values of $y^{(k)}$ to equation 4 and match the coefficients on each term of x^n so that they sum to zero and match the right hand side.

Example 4.1.1. Consider Airy's equation: $y'' - xy = 0$ near $x_0 = 0$.

Since we have

$$\begin{aligned} y &= \sum_{n \geq 0} a_n x^n \\ y'' &= \sum_{n \geq 2} a_n n(n-1) x^{n-2} = \sum_{n \geq 0} a_{n+2} (n+2)(n+1) x^n \\ xy &= \sum_{n \geq 0} a_n x^{n+1} = \sum_{n \geq 1} a_{n-1} x^n, \end{aligned}$$

we can find the coefficients of y'' and xy on x^n and make them add to 0.

$$\begin{aligned} x^0: \quad a_2 \cdot 2 \cdot 1 &= 0 \implies a_2 = 0 \\ x^n, n \geq 1: \quad a_{n+2}(n+2)(n+1) - a_{n-1} &= 0 \\ \implies a_{n+2} &= \frac{a_{n-1}}{(n+2)(n+1)}, n \geq 1 \\ \implies a_{m+3} &= \frac{a_m}{(m+3)(m+2)}, m = n-1 \geq 0. \end{aligned}$$

Hence, for any $i \in \{0, 1, 2\}$ and putting $m + 3 = 3k + i$, we have that

$$a_{3k+1} = \frac{a_{3k+i-3}}{(3k+i)(3k+i-1)}.$$

Suppose we are given c_k and b_0 and know that $b_k = c_k b_{k-1}$ for every $k \geq 1$. We get the relation that

$$\begin{aligned} b_k &= c_k [c_{k-1} b_{k-2}] \\ &= c_k c_{k-1} c_{k-2} b_{k-3} \\ &= c_k \cdots c_1 b_0, \end{aligned}$$

so we have the explicit formula

$$b_k = b_0 \prod_{i=1}^k c_i.$$

Put $b_k = a_{3k+i}$. Then

$$b_{k-1} = a_{3(k-1)+i} = a_{3k+i-3}.$$

Putting $c_k = \frac{1}{(3k+i)(3k+i-1)}$, we get that

$$b_k = b_{k-1} c_k.$$

Using the product relation from above,

$$a_{3k+i} = a_i \cdot \underbrace{\prod_{j=1}^k \frac{1}{(3j+1)(3j+i-1)}}_{A_k: i=0, B_k: i=1}.$$

We found earlier that $a_2 = 0$, so $a_{3k+2} = 0$. We now know that $a_{3k} = a_0 \cdot A_k$ and $a_{3k+1} = a_1 \cdot B_k$, where A_k, B_k are determined by the product.

Finally, we get the result

$$\begin{aligned} y(x) &= \sum_{i \geq 0} a_i x^i \\ &= a_0 \left(\sum_{k \geq 0} \underbrace{A_k x^{3k}}_{y_1(x)} \right) + a_1 \left(\sum_{k \geq 0} \underbrace{B_k x^{3k+1}}_{y_2(x)} \right), \end{aligned}$$

which is the general solution to Airy's equation (a_0, a_1 are free and modify y_1, y_2 , which are different).

Note that $|A_k|, |B_k| \leq 1$, so $\mu = \limsup_{n \rightarrow \infty} a_k^{\frac{1}{k}} \leq 1$. Thus, the power series converges if $|x - x_0| \leq \mu^{-1}$ and our computations are valid.

Example 4.1.2. Now suppose we have $x_0 = 1$ and the same equation of $y'' - xy = 0$. Then

$$\begin{aligned} y &= \sum_{n \geq 0} a_n (x-1)^n \\ y'' &= \sum_{n \geq 0} a_{n+2} (n+2)(n+1) (x-1)^n \\ xy &= \sum_{n \geq 0} a_n (x-1)^n + \sum_{n \geq 1} a_{n-1} (x-1)^n. \end{aligned}$$

Examine

$$\begin{aligned} (x-1)^0: \quad a_2 \cdot 2 \cdot 1 - a_0 &= 0 \\ (x-1)^n, \quad n \geq 1: \quad a_{n+2} (n+2)(n+1) - a_n - a_{n-1} &= 0 \\ \implies a_{n+2} &= \frac{a_n + a_{n-1}}{(n+2)(n+1)} \\ a_2 &= \frac{a_0}{2} \\ a_3 &= \frac{a_0 + a_1}{3 \cdot 2} \\ a_4 &= \frac{a_2 + a_1}{4 \cdot 3} = \frac{a_0}{2 \cdot 3 \cdot 4} + \frac{a_1}{3 \cdot 4} \end{aligned}$$

and so on — there is no clean solution for a_n in the form $a_n = A_n a_0 + B_n a_1^2$.

² I don't get this line

Theorem 4.1.3. Consider

$$P(x)y''(x) + Q(x)y'(x) + R(x)y(x) = 0 \quad (6)$$

with $P(x_0) \neq 0$, so we can write

$$y'' + \frac{Q}{P}y' + \frac{R}{P}y = 0. \quad (7)$$

If P, Q, R in (6) are analytic near x_0 with convergence radius R (respectively, $\frac{Q}{P}, \frac{R}{P}$ from (7)), then there exist power series solutions

$$y(x) = \sum_{n \geq 0} a_n (x - x_0)^n = a_0 y_1 + a_1 y_2$$

with convergence radius at least R .

Remark 4.1.4. We say P, Q, R have radius of convergence R where R is the minimum R such that

$$P(x) = \sum_{i \geq 0} p_i (x - x_0)^i, Q(x) = \sum_{i \geq 0} q_i (x - x_0)^i, R(x) = \sum_{i \geq 0} r_i (x - x_0)^i$$

for all x such that $|x - x_0| < R$.

Polynomials and exponential functions have $R = \infty$. $\ln$ and trigonometric functions are analytic (at least somewhere).

Nonsmooth functions like $Q = |x|^{\frac{1}{2}}$ are not analytic — requiring the coefficients in the ODE to be analytic is a very strong imposition.

4.2 Non-homogeneous Equations; Variation of Parameters

So far, we considered the right hand side of the ODE being zero, but now suppose we want to find the general solution to

$$Py'' + Qy' + Ry = g(x). \quad (8)$$

The key observation is that if ϕ, ψ solve (8), $\phi - \psi$ solves (6), so we can subtract to get

$$P(\phi - \psi)'' + Q(\phi - \psi)' + R(\phi - \psi) = 0,$$

which is a homogeneous equation.

Theorem 4.2.1. *Suppose that y_0 solves (8) and y_1, y_2 are two different solutions to (6). Then a general solution of (8) can be written as*

$$y(x) = y_0(x) + c_1y_1 + c_2y_2,$$

where c_1, c_2 are any real numbers.

Proof. Assume that y_1, y_2 are different solutions to (7), which normalizes our problem to

$$y'' + py' + qy = 0.$$

We want to find one solution to solve

$$y'' + py' + qy = g.$$

We conjecture the solution takes the form

$$y(x) = \mu_1(x)y_1 + \mu_2(x).$$

We introduced more unknowns to give ourselves more degrees of freedom. Computing, we find

$$\begin{aligned} y' &= y_1'\mu_1 + \mu_1 y_1' + \mu_2' y_2 + \mu_2 y_2' \\ y' &= \mu_1 y_1' + \mu_2 y_2', \end{aligned}$$

if we impose a constraint that

$$\mu_1' y_1 + \mu_2' y_2 = 0. \quad (9)$$

Continuing,

$$\begin{aligned} y'' &= \mu_1' y_1' + \mu_2' y_2' + \mu_1 y_1'' + \mu_2 y_2'' \\ py' &= p\mu_1 y_1' + p\mu_2 y_2' \\ qy &= q\mu_1 y_1 + q\mu_2 y_2 \\ \implies g &= y'' + py' + qy = \mu_1' y_1' + \mu_2' y_2' + \mu_1 \cdot 0 + \mu_2 \cdot 0, \end{aligned}$$

where the 0s in the last line come from y_1, y_2 being solutions to (7).

Hence, referring to our constraint (9) and the last line, we have the system

$$\begin{aligned}
 \mu'_1 y_1 + \mu'_2 y_2 &= 0 \\
 \mu'_1 y'_1 + \mu'_2 y'_2 &= g \\
 \Rightarrow \underbrace{\begin{pmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{pmatrix}}_A \begin{pmatrix} \mu'_1 \\ \mu'_2 \end{pmatrix} &= \begin{pmatrix} 0 \\ g \end{pmatrix} \\
 \Rightarrow \det(A) \neq 0 \text{ iff } y_1 \neq cy_2 \text{ or } y_2 \neq cy_1 \\
 \Rightarrow \begin{pmatrix} \mu'_1 \\ \mu'_2 \end{pmatrix} &= A^{-1} \begin{pmatrix} 0 \\ g \end{pmatrix}.
 \end{aligned}$$

□

5 Tuesday, October 14: Systems of First-Order Linear Equations

5.1 Basic Form

Suppose we have $a_n y^{(n)}(t) + a_{n-1} y^{(n-1)}(t) + \dots + a_1 y'(t) + a_0 y(t) = g(t)$. Rewriting this into a first-order system, we can put $x_1(t) = y(t)$, $x_2(t) = y'(t)$, and so on, so $y^{(n)} = \frac{d}{dt} x_n$.

We have

$$\begin{aligned}
 \frac{d}{dt} x_n &= -\frac{a_{n-1}}{a_n} x_n - \frac{a_{n-2}}{a_n} x_{n-1} - \dots - \frac{a_0}{a_n} x_1 + \frac{g(t)}{a_n} \\
 \frac{d}{dt} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} &= \begin{pmatrix} 0 & 1 & 0 & \dots \\ 0 & 0 & 1 & \dots \\ \vdots & & \ddots & \\ 0 & 0 & 0 & \dots \\ -\frac{a_0}{a_n} & \dots & \dots & -\frac{a_{n-1}}{a_n} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \vdots \\ \frac{g(t)}{a_n} \end{pmatrix}.
 \end{aligned}$$

In general, we have the unknowns $x_1(t), \dots, x_n(t)$ given $F_1, \dots, F_n$:

$$\frac{d}{dt} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} F_1(t, x_1, \dots, x_n) \\ \vdots \\ F_n(t, x_1, \dots, x_n) \end{pmatrix}.$$

Putting $x(t) = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ and $F = \begin{pmatrix} F_1 \\ \vdots \\ F_n \end{pmatrix}$, we have the general form

$$\frac{d}{dt} x(t) = F(t, x). \tag{10}$$

Note that (10) is linear iff

$$F(t, x) = A(t)x(t) + b(t)$$

for some matrix $A \in \mathbb{R}^{n \times n}$ and vector $b(t) \in \mathbb{R}^{n \times 1}$.

We get a first order linear system

$$\frac{d}{dt}x(t) = A(t)x(t) + b(t). \quad (11)$$

Recall that (11) is homogeneous if $b(t) = 0$ and nonhomogeneous otherwise. We will first investigate the homogeneous case.

Example 5.1.1. Consider the ODE $y'' + py' + qy = g(t)$.

Put

$$x_1 = 1$$

$$x_2 = y',$$

so

$$\begin{aligned} \frac{d}{dt}x_1 &= x_2 \\ \frac{d}{dt}x_2 &= -py' - qy + g = -px_2 - qx_1 + g. \end{aligned}$$

Hence

$$\frac{d}{dt} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ g \end{pmatrix}.$$

Given a general first order linear system as in (11), we are usually given an initial condition of $x(t_0) = x_0 \in \mathbb{R}^{n \times 1}$.

We will prove the following theorem next week.

Theorem 5.1.2. (Existence and Uniqueness) Suppose we have a first-order linear system as in (10) with an initial condition and $\left| \frac{\partial F_j}{\partial x_i} \right| \leq M$ and F_j is continuous for $t \in (a_0, b_0) = I_0$ and $x_i \in (a_i, b_i) = I_i$. In a certain range, F is Lipschitz continuous and continuous.

For any $t_0 \in I_0$ and $x_{0,i} \in I_i$, there exists $\delta > 0$ and a unique solution $x(t)$ for $t \in (t_0 - \delta, t_0 + \delta)$ with $|x(t) - x_0| < \delta$.

The conditions of the theorem are roughly equivalent to

$$|\nabla_x F| = |A(t)| \leq C \quad (12)$$

$$\implies A(t), b(t) \text{ are continuous.} \quad (13)$$

In particular, the partial derivatives of F need not be continuous.

5.2 The simplest case

Consider the linear and homogeneous system

$$\frac{d}{dt}x(t) = A(t)x(t). \quad (14)$$

What does a general solution look like? How many solutions do we have?

Lemma 5.2.1. Suppose we have the "first" solution $x^{(1)}(t) = \begin{pmatrix} x_1^{(1)} \\ \vdots \\ x_n^{(1)} \end{pmatrix}$ and

the "second" solution $x^{(2)}(t)$.

Then for any c_1, c_2 , $c_1x^{(1)} + c_2x^{(2)}$ solves (14).

Iterating, if $x^{(1)}, \dots, x^{(m)}$ solve the equation, then any $\sum_{i=1}^m c_i x^{(i)}$ solves (14) as well.

How do we distinguish different solutions?

Definition 5.2.2 (Linearly independent, Wronskian). Suppose that $x^{(1)}, \dots, x^{(m)}$ solve (14). We call $x^{(1)}, \dots, x^{(m)}$ linearly independent if the Wronskian (determinant) is nonzero; i.e.,

$$W(t) = \det \left(X^{(1)}, \dots, X^{(n)} \right) \neq 0$$

for all t .

Theorem 5.2.3. The derivative of the Wronskian is the trace of the coefficient matrix times the Wronskian:

$$\frac{d}{dt}W(t) = \text{tr}(A(t))W(t).$$

Proof. Write the matrix

$$\begin{aligned} M(t) &= \begin{pmatrix} x^{(1)} & \dots & x^{(n)} \end{pmatrix} \\ &= \begin{pmatrix} x_1^{(1)} & \dots & x_1^{(n)} \\ \vdots & & \vdots \\ x_n^{(1)} & \dots & x_n^{(n)} \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}. \end{aligned}$$

(Step 1) We have the following result:

$$\frac{d}{dt} \det \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = \det \begin{pmatrix} b'_1 \\ \vdots \\ b_n \end{pmatrix} + \dots + \det \begin{pmatrix} b_1 \\ \vdots \\ b'_n \end{pmatrix}.$$

This definition of linear independence is motivated by the previous lemma, since the redundant solution can be absorbed into the other solutions.

This is because of the following two equations:

$$\frac{d}{dt}(c_1(t) \cdots c_n(t)) = \sum_{k=1}^n c_1 \cdots c_{k-1} c'_k c_{k+1} \cdots c_n$$

$$\det \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & & \vdots \\ a_{n,1} & \cdots & a_{n,n} \end{pmatrix} = \sum_{i_1, \dots, i_n} a_{1,i_1} a_{2,i_2} \cdots a_{n,i_n} (-1)^{\sigma(i_1, \dots, i_n)}.$$

(Step 2) Using (14), we know

$$\frac{d}{dt} x^{(i)} = Ax^{(i)},$$

so we select one row and write

$$\frac{d}{dt} x_k^{(i)} = [Ax^{(i)}]_k = \sum_{j=1}^n A_{k,j} x_j^{(i)}.$$

Fixing a k , this is true for any i .

Then

$$\frac{d}{dt} (x_k^{(1)}, \dots, x_k^{(n)}) = \sum_{j=1}^n A_{k,j} (x_j^{(1)}, \dots, x_j^{(n)})$$

$$\frac{d}{dt} b_k = \sum_{j=1}^n A_{k,j} b_j.$$

(Step 3) Combining the previous two steps,

$$\begin{aligned}
 \frac{d}{dt}W(t) &= \sum_{k=1}^n \det \begin{pmatrix} b_1 \\ \vdots \\ b'_k \\ \vdots \\ b_n \end{pmatrix} \\
 &= \sum_{k=1}^n \det \begin{pmatrix} & & b_1 & & \\ & & \vdots & & \\ & & b_{k-1} & & \\ A_{k,1}b_1 + A_{k,2}b_2 + \dots + A_{k,n}b_n & & & & \\ & & \vdots & & \\ & & b_n & & \end{pmatrix} \\
 &= \sum_{k=1}^n \det \begin{pmatrix} b_1 \\ \vdots \\ A_{k,k}b_k \\ \vdots \\ b_n \end{pmatrix} \\
 &= \sum_{k=1}^n A_{k,k} \det \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \\
 &= \sum_{k=1}^n A_{k,k} W(t) \\
 &= \text{tr}(A)W(t).
 \end{aligned}$$

□

Corollary 5.2.4. *If $W(t_0) = 0$ for some $t_0 \in I$, then $W(t) = 0$ for every $t \in I$. Hence, if $W(t_0) \neq 0$ for some $t_0 \in I$, then $W(t) \neq 0$ for all $t \in I$.*

Proof. Note that

$$\begin{aligned}
 W(t) &= W(t_0) \exp \underbrace{\left[\int_{t_0}^t \text{tr}(A(s)) ds \right]}_{\mu(t)} \\
 \implies W(t_0) = 0 &\iff W(t) = 0, \quad \forall t \in I.
 \end{aligned}$$

□

Hence, to check that solutions are linearly independent, we just need to check the value of the Wronskain at any point t_0 .

Theorem 5.2.5. Suppose that $X^{(1)}, \dots, X^{(n)}$ are linearly independent. Then any solution $x(t)$ to (14) can be uniquely written as

$$x(t) = \sum_{i=1}^n c_i x^{(i)}(t).$$

In particular, once we have n linearly independent solutions, any other solution can be written as a linear combination of the already-found solutions.

Proof. For a given $c \in \mathbb{R}^n$, write a potential solution

$$y_c(t) = x(t) - \sum_{i=1}^n c_i x^{(i)}(t).$$

By linearity, $y_c(t)$ solves (14).

Fix t_0 . We want to find c_i such that

$$\underbrace{(x^{(1)}(t_0) \cdots x^{(n)}(t_0))}_{M(t_0)} \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = x(t_0).$$

Since $x^{(1)}, \dots, x^{(n)}$ are assumed to be linearly independent, the matrix on the left, $M(t_0)$, is invertible, so

$$\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = M(t_0)^{-1} x(t_0) \\ \implies y_c(t_0) = 0.$$

Now that we know $y_c(t)$ solves (14) and that $y_c(t_0) = 0$, 0 is a solution to (14). By uniqueness, $y_c(t) = 0$. Going to the definition of y ,

$$x(t) = \sum_{i=1}^n c_i x^{(i)}(t).$$

□

6 Thursday, October 16: Systems of First Order Linear Equations, Cont'd

6.1 Dimension of the Solution Space, Linear Independence

Suppose we have the system

$$\frac{d}{dt}x(t) = A(t)x(t) \tag{15}$$

Recall from last time that for an n th order ODE, the solution space is spanned by an n -dimensional basis of solutions. If we find n linearly independent solutions, any real solution can be written as a linear combination of these solutions.

Theorem 6.1.1. Suppose that $X^{(i)}$ solves

$$\frac{d}{dt}X^{(i)} = A(t)X^{(i)}$$

and the initial condition satisfies $X^{(i)}(t_0) = e_i$ for each $i \in \{1, \dots, n\}$. Then $X^{(1)}, \dots, X^{(n)}$ are linearly independent real-valued solutions.

Proof. By the existence-uniqueness theorem, we know that $X^{(i)}(t)$ exists. Furthermore, for $W(t) = \det(X^{(1)} \cdots X^{(n)}) \neq 0$ for every $t \in I$, since $(X^{(1)} \cdots X^{(n)})$ is just the n -dimension identity matrix,

$$\begin{aligned} \frac{d}{dt}W(t) &= \text{tr}(A)W(t) \\ W(t_0) \neq 0 &\iff W(t) \neq 0 \text{ for every } t \in I. \end{aligned}$$

At t_0 , $W(t_0) = \det(I_n) \neq 0$, so the solutions are linearly independent. $\square$

The last theorem from last class says that the dimension of the solution space is at most n . Today's theorem says that the dimension is at least n .

Example 6.1.2. ($n = 2$) Suppose we have $y'' + py' + qy = 0$. We can turn this into a 2×2 first-order linear system. We know that the dimension of the solution space is 2.

6.2 Finding Linearly Independent Solutions

For intuition, consider that when $n = 1$, we have $\frac{d}{dt}x(t) = a(t)x(t)$.

The solution is $x(t) = x(t_0)e^{\int_{t_0}^t a(s)ds}$

For $n \geq 2$, in a special case when $A(t)$ is diagonal,

$$\begin{aligned} \frac{d}{dt}x &= Ax \\ \frac{d}{dt} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} &= \begin{pmatrix} \ddots & & \\ & \ddots & \\ & & \ddots \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} A_{11}x_1 \\ \vdots \\ A_{nn}x_n \end{pmatrix}. \end{aligned}$$

We get n ODEs, where $\frac{d}{dt}x_i = A_{ii}x_i$ for $i \in \{1, \dots, n\}$. This is only simple when A is constant.

Suppose $A \in \mathbb{R}^{n \times n}$ is constant and we want to solve

$$\frac{d}{dt}x = Ax.$$

We conjecture the solution takes the form $x(t) = e^{\lambda t}v$ for $v \in \mathbb{R}^{n \times 1}$.

As an analog, $y'' + p(t)y' + q(t)y = 0$ must be solved using power series (complicated) when p, q are not constant. Even when p, q are constant, the solution can still be difficult to find.

Differentiating,

$$\begin{aligned}\frac{d}{dt}(e^{\lambda t}v) &= \lambda e^{\lambda t}v \\ &= t e^{\lambda t}v = \\ &e^{\lambda t}Av.\end{aligned}$$

This equality is true iff $\lambda v = Av$.

In this n -dimensional space, we can project into the eigendirection and we get 1×1 ODEs (we have one first-order ODE).

Our goal is to find n different real solutions. There are several cases to examine:

- (i) There are n distinct real eigenvalues λ_i .
- (ii) There are some complex eigenvalues.
- (iii) There are repeated eigenvalues.
- (iv) (There are n real linearly independent eigenvectors)

The fourth case is contained in the others.

(Case 1) Suppose we have n distinct real eigenvalues λ_i . Then

$$\det(A - \lambda_i I) = 0,$$

so

$$Av_i = \lambda_i v_i$$

for some $v_i \in \mathbb{R}^{n \times 1}$.

For any $i \in \{1, \dots, n\}$, we can put $x^{(i)} = e^{\lambda_i t} v_i$ to generate n solutions³. There is an associated results that if $\lambda_1, \dots, \lambda_n$ are different, then $v_1, \dots, v_n$ are linearly independent.

³ find reference

Linear independence gives us $W(0) = \det(v_1 \cdots v_n) \neq 0$, so we have found n solutions. This implies that case 1 is a strict subset of case 4.

(Case 4) If we assume that v_i are real, $Av_i = \lambda_i v_i$ implies that λ_i is real.

$$\begin{aligned}Av &= (\lambda_1 v_1, \dots, \lambda_n v_n) \\ &= \underbrace{(v_1, \dots, v_n)}_P \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}.\end{aligned}$$

v_i being linearly independent gives us that P is invertible where

$$\begin{aligned}AP &= P\Lambda, \quad \Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \\ \iff A &= P\Lambda P^{-1}.\end{aligned}$$

So A is diagonalizable in case 4.

Theorem 6.2.1. (Symmetric) If $A = A^T \in \mathbb{R}^{n \times n}$, then $A = Q\Lambda Q^T$ for $QQ^T = Q^TQ = I_n$, so $Q^{-1} = Q^T$. Furthermore, $\lambda_i \in \mathbb{R}$ for $i \in \{1, \dots, n\}$.

In other words, if we have a symmetric matrix, then the ODE can be solved in the way of case 4, with $x^{(i)}(t) = e^{\lambda_i t} v_i$. Since $W(0) = \det(v_1 \cdots v_n) \neq 0$, we are done.

(Case 2) Since A is real, we have that

$$\begin{aligned} Av &= \lambda v \\ A\bar{v} &= \bar{\lambda}\bar{v}, \end{aligned}$$

so having one solution $x^{(1)}(t) = e^{\lambda t} v$ gives us another solution $x^{(2)}(t) = e^{\bar{\lambda} t} \bar{v}$. These are two complex solutions; we want to derive two real solutions from them.

Suppose $\lambda = \alpha + i\beta$ and $v = a + ib$. Then after expanding,

$$\begin{aligned} x(t) &= e^{\lambda t} v = e^{(\alpha + i\beta)t} (a + ib) \\ &= e^{-\alpha t} [a \cos(\beta t) - b \sin(\beta t)] + i [a \sin(\beta t) + b \cos(\beta t)] \\ \bar{x}(t) &= e^{-\alpha t} [a \cos(\beta t) - b \sin(\beta t)] + i [a \sin(\beta t) + b \cos(\beta t)]. \end{aligned}$$

Since $x(t), \bar{x}(t)$ solve (15),

$$\operatorname{Re} x(t) = \frac{x(t) + \bar{x}(t)}{2}, \quad \operatorname{Im} x(t) = \frac{x(t) - \bar{x}(t)}{2i}.$$

These two real equations solve the ODE since $x(t) = c(t) + id(t), \bar{x}(t) = c(t) - id(t)$ means that c, d are solutions. In summary, we took a pair of eigenvector-value pairs, got two complex solutions, then took the real and imaginary components for two real solutions.

$e^{\lambda t} v$ can solve cases 1, 2, and 4, so long as there are no repeated eigenvalues.

7 Tuesday, October 21: Systems of First-Order Linear Equations, Cont'd.

7.1 Review

Recall to solve (for x a vector)

$$\frac{d}{dt} x = Ax,$$

we get $x(t) = e^{\lambda t} v$ for $Av = \lambda v$. Last time, we considered when v were linearly independent and real. If v are complex eigenvectors, then we can take the real and imaginary parts as solutions. Today, we examine when there are repeated roots, so there are solutions not in the form $x(t) = e^{\lambda t} v$.

7.2 Matrix Exponential

($n = 1$) If we have $\frac{d}{dt}x = ax$ for $a \in \mathbb{R}$, then the solution is

$$x(t) = e^{at}x_0.$$

($n \geq 2$) We conjecture the solution looks like

$$x(t) = e^{At}x_0$$

for some matrix exponential operator that satisfies $\frac{d}{dt}e^{At} = Ae^{At}$.

Definition 7.2.1 (Matrix exponential). Define the matrix exponential operator by

$$\exp[A] = \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

If the sum converges, we get a matrix in the same dimension ($n \times n$) as A .

Example 7.2.2. Suppose A is diagonal, so $A = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$. Then

$$A^k = \begin{pmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{pmatrix},$$

so

$$\exp[A] = \begin{pmatrix} \sum_{k=0}^{\infty} \frac{\lambda_1^k}{k!} & & \\ & \ddots & \\ & & \sum_{k=0}^{\infty} \frac{\lambda_n^k}{k!} \end{pmatrix} = \begin{pmatrix} e^{\lambda_1} & & \\ & \ddots & \\ & & e^{\lambda_n} \end{pmatrix}.$$

Example 7.2.3. Let $A = \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix} = \beta J$, $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Then

$$A^k = \beta^k J^k.$$

We see that $J^2 = -I$, $J^3 = -J$, $J^4 = I$, so

$$\begin{aligned} A^{4k+i} &= \beta^{4k+i} J^{4k+i} \\ &= \beta^{4k+1} J^i. \end{aligned}$$

We can show that

$$\exp[A] = \sum_{k=0}^{\infty} \frac{A^k}{k!} = \begin{pmatrix} D & C \\ -C & D \end{pmatrix} = \begin{pmatrix} \cos(\beta) & \sin(\beta) \\ -\sin(\beta) & \cos(\beta) \end{pmatrix}.$$

Where is the matrix exponential well-defined? Where does it converge? If

$$(\exp[A])_{ij} = \sum_{k \geq 0} \frac{(A^k)_{ij}}{k!}$$

converges for every i, j , then the matrix exponential converges.

We want to control the growth rate as A is multiplied against itself. Suppose $\max_{ij} |A_{ij}| \leq a$ for $A \in \mathbb{R}^{n \times n}$. We want to prove that $|(A^k)_{ij}| \leq (na)^{k-1} a$ through induction. In $k = 1$, the result is trivial.

Suppose that $\max_{ij} |(A^k)_{ij}| \leq (na)^{k-1} a$ for some k . Then

$$\begin{aligned} (A^{k+1})_{ij} &= (A^k A)_{ij} \\ &= \sum_{i=1}^n [(A^k)_{ij} \cdot A_{ij}] \\ \implies |(A^{k+1})_{ij}| &\leq \left| \sum_{i=1}^n (na)^{k-1} a \cdot a \right| \\ &= (na)^k a. \end{aligned}$$

Consequently,

$$\begin{aligned} \sum_{k=0}^N \frac{|(A^k)_{ij}|}{k!} &\leq \sum_{k=0}^{\infty} \frac{(na)^{k-1} a}{k!} \\ &= \frac{1}{n} \sum_{k=0}^{\infty} \frac{(na)^k}{k!} \\ &= \frac{1}{n} e^{na} < \infty. \end{aligned}$$

So the matrix exponential is well-defined for every $A \in \mathbb{R}^{n \times n}$.

Lemma 7.2.4. (i) Suppose $B = T^{-1}AT$. Then

$$\exp[B] = T^{-1} \exp[A] T.$$

(ii) If $AB = BA$, then $\exp[A + B] = \exp[A] \exp[B]$

(iii) $(\exp[A])^{-1} = \exp[-A]$

Proof. (i) Note that $B^k = (T^{-1}AT)(T^{-1}AT) \cdots (T^{-1}AT) = T^{-1}A^kT$. Then

$$\begin{aligned} \sum_{k \geq 0} \frac{B^k}{k!} &= \sum_{k=0}^{\infty} T^{-1} \frac{A^k}{k!} T \\ &= T^{-1} \left(\sum_{k=0}^{\infty} \frac{A^k}{k!} \right) T \\ &= T^{-1} \exp[A] T. \end{aligned}$$

(ii) Compute

$$\begin{aligned}
 \exp[A + B] &= \sum_{k=0}^{\infty} \frac{(A + B)^k}{k!} \\
 &= \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{i=0}^k A^i B^{k-i} \binom{k}{i} = \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{i=0}^k A^i B^{k-i} \frac{k!}{i!(k-i)!} \\
 &= \sum_{k=0}^{\infty} \sum_{i=0}^k A^i B^{k-i} \frac{1}{i!(k-i)!} \\
 &= \sum_{i=0}^{\infty} \sum_{k=i}^{\infty} \frac{A^i}{i!} \frac{B^{k-i}}{(k-i)!} \\
 \text{putting } k &= i + j, = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \frac{A^i}{i!} \frac{B^j}{j!} \\
 &= \sum_{i=0}^{\infty} \frac{A^i}{i!} \sum_{j=0}^{\infty} \frac{B^j}{j!} \\
 &= \exp[A] \exp[B].
 \end{aligned}$$

The fourth equality follows from writing out what pairs of (i, k) are hit by the double sum. The binomial substitution follows from the assumption that $AB = BA$.

(iii) Apply (ii) with $B = -A$. Note that $AB = BA = -A^2$ and $\exp[A + B] = \exp[0 \cdot I] = I$. Then **not sure**

□

Proposition 7.2.5. Suppose $Av = \lambda v$. Then $\exp[A]v = e^{\lambda}v$.

Proof. Left as an exercise; use $A^k v = \lambda^k v$.

□

Proposition 7.2.6. Applying the derivative component-wise, $\frac{d}{dt} \exp[tA] = A \exp[tA] = \exp[tA]A$.

Proof. Compute

$$\begin{aligned}
 \frac{d}{dt} \left(\underbrace{\sum_{k=0}^{\infty} \frac{(tA)^k}{k!}}_{\leq \frac{1}{n} e^{nat}} \right) &= \sum_{k=0}^{\infty} \frac{d}{dt} \frac{(tA)^k}{k!} \\
 &= \sum_{k=1}^{\infty} \frac{A^k}{k!} k t^{k-1} \\
 &= \sum_{k=1}^{\infty} \frac{A^k}{(k-1)!} t^{k-1} \\
 &= A \sum_{k=1}^{\infty} \frac{(At)^{k-1}}{(k-1)!} \\
 &= A \exp[At] = \exp[At]A,
 \end{aligned}$$

where the last equality follows because the powers in the third to last line can commute. $\square$

Theorem 7.2.7. Suppose that $A \in \mathbb{R}^{n \times n}$. Then the solution to

$$\frac{d}{dt}x = Ax \quad (16)$$

with the initial condition $x(t_0) = x_0 \in \mathbb{R}^n$ is given by

$$x(t) = e^{A(t-t_0)}x_0.$$

As we hoped, this is very similar to the scalar case, where $a \in \mathbb{R}$ so $x(t) = e^{at}x_0$.

Proof. Define $x(t) = e^{A(t-t_0)}x_0$. Compute

$$\begin{aligned} \frac{d}{dt}x(t) &= \frac{d}{dt}e^{A(t-t_0)}x_0 \\ &= Ae^{A(t-t_0)}x_0 \quad \text{by (ii)} \\ &= Ax(t). \end{aligned}$$

Note that

$$x(t_0) = e^{A \cdot 0}x_0 = x_0,$$

so x satisfies (16) and the initial condition. $\square$

Suppose that we pick the initial condition as an eigenvector of A , so $x_0 = v$ and $Av = \lambda v$. By Proposition 7.2.5, we get that

$$x(t) = e^{At}v = e^{\lambda t}v.$$

This shows that the matrix exponential method covers the solution method we used for linearly independent eigenvectors.

Summarizing,

- (i) Case 1: we have n distinct real eigenvalues
- (ii) Case 2: we have n distinct complex eigenvalues, and can use the real and imaginary components to get n linearly independent real eigenvectors.

In both of these cases, if we have $Av_i = \lambda_i v_i$ for linearly independent v_i , then

$$\begin{aligned} A \left(\underbrace{v_1 v_2 \cdots v_n}_P \right) &= (v_1 v_2 \cdots v_n) \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \\ AP &= P\Lambda \\ \implies A &= P\Lambda P^{-1}, \end{aligned}$$

If A is Lipschitz, then the solution is unique. But we have been considering A as a constant matrix. We call e^{At} the fundamental matrix.

so A is diagonalizable by P , where $\Lambda \in \mathbb{C}^{n \times n}$. Here we have

$$\begin{aligned} e^{At} &= P e^{\Lambda t} P^{-1} \\ &= P \begin{pmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_n t} \end{pmatrix} P^{-1}. \end{aligned}$$

(i) Case 3: We have repeated roots.

Theorem 7.2.8. (Jordan form) Suppose that $A \in \mathbb{C}^{n \times n}$. Then there exists $U, J \in \mathbb{C}^{n \times n}$ such that

$$A = UJU^{-1}, \quad J = \begin{pmatrix} J_1 & & \\ & \ddots & \\ & & J_m \end{pmatrix},$$

and

$$J_i = \begin{pmatrix} \alpha_i & 1 & & \\ & \ddots & \ddots & 1 \\ & & \ddots & \\ & & & \alpha_i \end{pmatrix} \in \mathbb{C}^{n_i \times n_i},$$

where α_i are the eigenvalues of A and $n_1 + n_2 + \dots + m = n$. $U = (u_1 \cdots u_n)$, where u_i are generalized eigenvectors.

$$(A - \lambda I)^k U_i = 0.$$

By Lemma 7.2.4, we have that for $A = UJU^{-1}$

$$e^{At} = U \exp[Jt] U^{-1},$$

Because $\exp[Jt] = \begin{pmatrix} e^{J_1 t} & & \\ & \ddots & \\ & & e^{J_m t} \end{pmatrix}$ and $J_i = \alpha_i I + N$,

N is the matrix with an offset diagonal.

$$e^{J_i t} = e^{\alpha_i t I + N t} = \exp(\alpha_i t I) \exp(N t).$$

$$\exp[\alpha_i t I] = e^{\alpha_i t} I$$

and

$$\exp[Nt] = I + Nt + \frac{N^2 t^2}{2} + \dots + \frac{N^{n_i-1} t^{n_i-1}}{(n_i-1)!} = \begin{pmatrix} 1 & t & \frac{t^2}{2} & \cdots & \frac{t^{n_i-1}}{(n_i-1)!} \\ 0 & 1 & t & \cdots & \\ & & 1 & \cdots & \\ \ddots & & & \ddots & \\ & & & & 1 \end{pmatrix}.$$

As an example, when

$$\frac{d}{dt} X = \begin{pmatrix} \lambda & 1 \\ & \lambda & 1 \\ & & \lambda \end{pmatrix},$$

we get

$$e^{At} = e^{\lambda t} e^{Nt} = e^{\lambda t} \begin{pmatrix} 1 & t & \frac{t^2}{2} \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}.$$

8 Thursday, October 24: Non-homogeneous equations, Existence-Uniqueness Theorem

8.1 Non-homogeneous equations

Suppose we want to solve a nonhomogeneous system, where for a constant matrix $A \in \mathbb{R}^{n \times n}$ and $x \in \mathbb{R}^n$, we are given

$$\frac{d}{dt}x = Ax + G(t) \quad (17)$$

$$x(t_0) = x_0. \quad (18)$$

($n = 1$) Suppose we have $\frac{d}{dt}x = ax + g(t)$ with the initial condition $x(t_0) = x_0$. Use an integrating factor of e^{-at} to get

$$\begin{aligned} \frac{d}{dt}(e^{-at}x(t)) &= (-ax(t) + \dot{x}(t))e^{-at} \\ &= g(t)e^{-at} \\ \implies e^{-at}x(t) &= e^{-at_0}x(t_0) + \int_{t_0}^t g(s)e^{-as}ds \\ \implies x(t) &= e^{a(t-t_0)}x_0 + \int_{t_0}^t e^{a(t-s)}g(s)ds \\ x(t) &= e^{a(t-t_0)} \left[x_0 + \int_{t_0}^t e^{-a(s-t_0)}g(s)ds \right]. \end{aligned}$$

Note that $e^{a(t-t_0)}x_0$ solves

$$\frac{d}{dt}x_1 = ax_1, \quad x_1|_{t=t_0} = x_0,$$

while $\int_{t_0}^t g(s)e^{-as}ds$ solves

$$\frac{d}{dt}x_2 = ax_2 + g(t), \quad x_2(t_0) = 0.$$

This gives us intuition for the $n \geq 2$ case. e^a will turn into e^A , while g will turn into G .

Theorem 8.1.1. *The solution to (17) is given by*

$$\begin{aligned} x(t) &= e^{A(t-t_0)}x_0 + \int_{t_0}^t e^{A(t-s)}G(s)ds \\ &= e^{A(t-t_0)} \left(x_0 + \int_{t_0}^t e^{-A(s-t_0)}G(s)ds \right). \end{aligned}$$

Proof. Put $c(t) = x_0 + \int_{t_0}^t e^{-A(s-t_0)} G(s) ds$ and $x(t) = e^{A(t-t_0)} c(t)$. We first want to verify that $x(t)$ solves (17), as by uniqueness, it will be the only solution.

We can compute

$$\begin{aligned} \frac{d}{dt} x(t) &= A e^{A(t-t_0)} c(t) + e^{A(t-t_0)} \dot{c}(t) \\ &= A x + e^{A(t-t_0)} e^{-A(s-t_0)} G(s) \Big|_{s=t} \\ &= A x + e^{A(t-t_0)} e^{-A(t-t_0)} G(t) \\ &= A x + G(t). \end{aligned}$$

□

To summarize our solution method to linear systems of first-order ODEs with a constant matrix A , with $\exp(tA)$, we are able to solve the homogeneous problem. We solve nonhomogeneous systems with the formula given by the above theorem.

A really just has to be independent of t .

8.2 Existence-Uniqueness Theorem

Theorem 8.2.1. (*Existence-Uniqueness Theorem*) Suppose

$$\frac{d}{dt} y = f(t, y(t)) \in \mathbb{R}, y(t_0) = y_0. \quad (19)$$

Define a region $R : |t - t_0| \leq a, |y - y_0| \leq b$.

Assume the following hold:

- (i) $f(t, y(t))$ is continuous on R ,
- (ii) $f(t, y(t))$ is Lipschitz in y on R , so

$$|f(t, y_1) - f(t, y_2)| \leq L |y_1 - y_2|$$

for any $(t_1, y_1), (t_2, y_2) \in R$. Note that if $\frac{\partial f}{\partial y}$ exists and is continuous, then this is satisfied. (Continuity of the partial is a stronger assumption; e.g. $|y|$.)

Then there exists $h > 0$ such that (19) admits a unique C^1 solution for $|t - t_0| \leq h$.

The proof and theorem generalize to $n \geq 2$ by replacing $|y|$ with $\|y\|$.

The easiest guess is $\phi(t) = y_0$, i.e. $\phi(t) = y_0 + o(1)$ as $t \rightarrow t_0$. The next easiest guess is the first-order Taylor approximation

$$\phi(t) = y_0 + y'(t_0)(t - t_0) = y_0 + f(t_0, y_0)(t - t_0) + o(|t - t_0|),$$

which actually satisfies the slope condition at t_0 . If f were sufficiently smooth, we could continue this process: since $y'(t) = f(t, y(t))$, we would have

$$y^{(n+1)}(t) = \frac{d^n}{dt^n} f(t, y(t)),$$

and hence obtain higher-order Taylor terms for y about t_0 . We cannot justify that expansion here, though, because we only assumed that f is continuous (in t) and Lipschitz in y .

What if we try putting $\phi(t) = y_0 + \int_{t_0}^t f(s, \phi(s)) ds$, another function where $\frac{d}{dt} \phi(t) = f(t, \phi(t))$? We can get

$$\begin{aligned}\phi_0(t) &= y_0 \\ \phi_1(t) &= y_0 + \int_{t_0}^t f(s, \phi_0(s)) ds \\ \phi_2(t) &= y_0 + \int_{t_0}^t f(s, \phi_1(s)) ds \\ \phi_{n+1}(t) &= y_0 + \int_{t_0}^t f(s, \phi_n(s)) ds\end{aligned}$$

and so on. We would hope that this converges to an actual solution when $|t - t_0|$ is small. This method is called *Picard iteration*.

Proof. (Step 1) We want to make sure $|\phi_n(t) - y_0| \leq b$ for $|t - t_0| \leq a$. Assumption (i) implies that $|f| \leq M$ in R . We then know

$$|\phi_{n+1} - y_0| \leq \int_{t_0}^t |f| \leq M |t - t_0|.$$

Then if $|t - t_0| \leq h \leq \frac{b}{M}$, $|\phi_{n+1}(t) - y_0| \leq b$. Our region determined by h is constant among iterations.

(Step 2) We want to show $|\phi_{n+1} - \phi_n| \rightarrow 0$ as $n \rightarrow \infty$.

$$\begin{aligned}\phi_{n+1}(t) &= y_0 + \int_{t_0}^t f(s, \phi_n(s)) ds \\ \phi_n(t) &= y_0 + \int_{t_0}^t f(s, \phi_{n-1}(s)) ds \\ \implies (\phi_{n+1} - \phi_n)(t) &= \int_{t_0}^t (f(s, \phi_n(s)) - f(s, \phi_{n-1}(s))) ds.\end{aligned}$$

By assumption (ii) and the triangle inequality,

$$|\phi_{n+1} - \phi_n| \leq \int_{t_0}^t L |\phi_n - \phi_{n-1}(s)| ds.$$

To iterate backwards, we iterate forwards.

$$\begin{aligned}|(\phi_1 - \phi_0)(t)| &= |\phi_1 - y_0| \leq Mh, \quad |t - t_0| < h \\ |\phi_2 - \phi_1| &\leq \int_{t_0}^t L |\phi_1 - \phi_0| ds \\ &\leq L |t - t_0| Mh \leq LhMh\end{aligned}$$

inducting, we see that $|(\phi_{n+1} - \phi_n)(t)| \leq (Lh)^n Mh$.

Hence, the requirement for h is that $h < \min\{\frac{b}{M}, \frac{1}{L}, 1\}$.

$h \leq \frac{b}{M}$ can bind, but $h < \frac{1}{L}$ must hold strictly.

(Step 3) We want to show convergence in $\phi_n(t) \rightarrow \phi(t)$. To do so, we need to first show that ϕ is well defined for $t \in [t_0 - h, t_0 + h]$.

$$\begin{aligned}\phi(t) &= \lim_{n \rightarrow \infty} \phi_{n+1} - \phi_0 + \phi_0 \\ &= \lim_{n \rightarrow \infty} \sum_{i=0}^n (\phi_{i+1} - \phi_i) + \phi_0.\end{aligned}$$

By step 2, we see that

$$\begin{aligned}\sum_{i=0}^{\infty} |\phi_{i+1} - \phi_i| &\leq \sum_{i=0}^{\infty} (Lh)^i Mh \\ &= \frac{1}{1 - Lh} Mh < \infty.\end{aligned}$$

so the series $\sum_{i=0}^n (\phi_{i+1} - \phi_i)$ converges absolutely, so it converges for all t where $|t - t_0| \leq h$. Hence, $\phi(t)$ is well defined on this interval.

Now we write by the triangle inequality

$$\begin{aligned}|\phi(t) - \phi_n(t)| &\leq \sum_{i=n}^{\infty} |\phi_{i+1} - \phi_i| \\ &\leq \sum_{i=n}^{\infty} (Lh)^i Mh \rightarrow 0,\end{aligned}$$

so $\phi_n \rightarrow \phi$ uniformly.

(Step 4) We want to show that ϕ solves $\phi(t) = y_0 + \int_{t_0}^t f(s, \phi(s)) ds$. We write

$$\phi(t) = \lim_{n \rightarrow \infty} \phi_n(t) = \lim_{n \rightarrow \infty} y_0 + \int_{t_0}^t f(s, \phi_n(s)) ds$$

and pass the limit through the integral by the Dominated Convergence Theorem, then pass the limit through f by continuity of f .

Furthermore,

$$\frac{d}{dt} \phi(t) = f(t, \phi(t)),$$

and since f was assumed to be continuous, ϕ is C^1 . So the limit of our sequence is a solution to the ODE.

(Step 5) Now we show uniqueness. Suppose ϕ and ψ both solve

$$\phi(t) = y_0 + \int_{t_0}^t f(s, \phi(s)) ds, \quad \psi(t) = y_0 + \int_{t_0}^t f(s, \psi(s)) ds.$$

By the Lipschitz condition in x with constant L ,

$$\begin{aligned}|\phi(t) - \psi(t)| &= \left| \int_{t_0}^t (f(s, \phi(s)) - f(s, \psi(s))) ds \right| \\ &\leq \int_{t_0}^t L |\phi(s) - \psi(s)| ds = L \int_{t_0}^t |\phi(s) - \psi(s)| ds.\end{aligned}$$

Define the nonnegative function

$$D(t) := \int_{t_0}^t |\phi(s) - \psi(s)| ds, \quad D(t_0) = 0.$$

Then $D'(t) = |\phi(t) - \psi(t)| \leq LD(t)$. Hence

$$\frac{d}{dt}(e^{-Lt}D(t)) = e^{-Lt}(D'(t) - LD(t)) \leq 0 \Rightarrow e^{-Lt}D(t) \leq e^{-Lt_0}D(t_0) = 0.$$

Since $e^{-Lt} > 0$ and $D(t) \geq 0$, we conclude $D(t) = 0$ for all t in the interval, so $|\phi(t) - \psi(t)| = D'(t) = 0$ and therefore $\phi = \psi$. $\square$

Remark 8.2.2. Assumption (i), that $f \in C^0$ gave us existence. Assumption (ii), that f is Lipschitz continuous, gave us uniqueness.

9 Tuesday, October 28: Quantitative Estimates

Last time, we showed that if f is continuous in a rectangle around $t, x(t)$, then there exists a solution to $\frac{d}{dt}x = f(t, x)$. If f is Lipschitz in x , then there is a unique solution.

9.1 Gronwall's Inequality

Lemma 9.1.1. Suppose α, β are given and we know that

$$\psi(t) \leq \alpha(t) + \int_0^t \beta(s)\psi(s)ds \quad (20)$$

$$\beta(t) \geq 0. \quad (21)$$

Define

$$B(s) = \exp\left(\int_0^s \beta(t)dt\right).$$

Then

$$\psi(t) \leq \alpha(t) + \int_0^t \alpha(s)\beta(s) \frac{B(t)}{B(s)} ds,$$

which we can integrate by parts to find that

$$\psi(t) \leq \alpha(0)B(t) + \int_0^t \alpha'(s) \frac{B(t)}{B(s)} ds.$$

Remark 9.1.2. Once we prove this, we will have the following applications:

(i) Our motivation is from last class: we used that

$$d(t) := |\phi(t) - \psi(t)| \leq L \int_{t_0}^t |\phi(s) - \psi(s)| ds = L \int_{t_0}^t d(s) ds.$$

Gronwall's inequality is a more general version of this. If we take $\alpha = 0$ and $\beta = L$, then we recover $d \leq 0$.

(ii) If $\alpha = 0$, then $\psi \leq 0$ automatically.

(iii) We also have the application that if $\alpha(t) = \alpha_0 + \alpha_1 t$ and $\beta = L$, we get

$$\begin{aligned} B(s) &= \exp\left(\int_0^s L ds\right) = e^{Ls} \\ \implies \psi(t) &\leq \alpha_0 e^{Lt} + \int_0^t \alpha_1 e^{L(t-s)} ds \end{aligned}$$

by the post-IBP form of Gronwall's inequality. The result is Duhamel's formula.

(iv) Hence, if $\alpha = A$ and $\beta = L$ are constants, then

$$\psi(t) \leq Ae^{Lt}.$$

Proof. We want to rewrite the givens into an ODE inequality. Introduce

$$A(t) = \int_0^t \beta(s)\psi(s)ds.$$

We want to solve an ODE inequality for A .

Taking a derivative of A , we get

$$A'(t) = \beta(t)\psi(t).$$

Using the first assumption (20), we get

$$\psi(t) \leq \alpha(t) + A(t),$$

and combining this with the second assumption (21), we get

$$\frac{d}{dt}A(t) = \psi(t)\beta(t) \leq \alpha(t)\beta(t) + A(t)\beta(t).$$

We use an integrating factor to solve

$$A'(t) - \beta(t)A(t) \leq \alpha(t)\beta(t).$$

Since

$$B(t) = \exp\left(\int_0^t \beta(s)ds\right),$$

we see

$$B^{-1}(t) = \exp\left(\int_0^t -\beta(s)ds\right)$$

so that

$$\frac{d}{dt}B^{-1}(t) = -\beta(t)B^{-1}(t).$$

Then

$$\begin{aligned} \frac{d}{dt}\left[A(t)B^{-1}(t)\right] &= A'(t)B^{-1}(t) + A(t)\left(B^{-1}\right)'(t) \\ &= A'(t)B^{-1}(t) - \beta(t)A(t)B^{-1}(t) \\ &= (A'(t) - \beta(t)A(t))B^{-1}(t) \\ &\leq \alpha(t)\beta(t)B^{-1}(t). \end{aligned}$$

Duhamel's formula says that the solution to $\frac{d}{dt}x = Lx + \alpha_1$ with the initial condition $x(0) = \alpha_0$ is given by

$$x(t) = e^{Lt}\alpha_0 + \alpha_1 \int_0^t e^{L(t-s)} ds.$$

Solving an ODE inequality is taking $\frac{d}{dt}y = f(t, y(t))$ and ending with

$$\frac{d}{dt}[\text{unknown}] \leq \text{known}.$$

We can integrate to get an upper bound on the unknown.

Be cautious: while $f \leq g$ implies $\int f \leq \int g$, it does *not* imply that $\dot{f} \leq \dot{g}$.

We can rewrite the final integral using integration by parts. First note that

$$B'(s) = \beta(s)B(s) \implies \frac{d}{ds}\left(\frac{B(t)}{B(s)}\right) = -\beta(s)\frac{B(t)}{B(s)}.$$

Using integration by parts with

$$u(s) = \alpha(s), dv(s) = \beta(s)\frac{B(t)}{B(s)}ds = -d\left(\frac{B(t)}{B(s)}\right),$$

we have $du = \alpha'(s)$, $v = -\frac{B(t)}{B(s)}$, and

$$\begin{aligned} &\int_0^t \alpha(s)\beta(s)\frac{B(t)}{B(s)}ds \\ &= \int_0^t u(s)dv(s) \\ &= u(s)v(s)\Big|_0^t - \int_0^t u'(s)v(s)ds \\ &= -\alpha(s)\frac{B(t)}{B(s)}\Big|_0^t + \int_0^t \alpha'(s)\frac{B(t)}{B(s)}ds \\ &= -\alpha(t)\frac{B(t)}{B(t)} + \alpha(0)\frac{B(t)}{B(0)} + \int_0^t \alpha'(s)\frac{B(t)}{B(s)}ds \\ &= -\alpha(t) + \alpha(0)B(t) + \int_0^t \alpha'(s)\frac{B(t)}{B(s)}ds. \end{aligned}$$

Substituting this into

$$\psi(t) \leq \alpha(t) + \int_0^t \alpha(s)\beta(s)\frac{B(t)}{B(s)}ds,$$

we obtain

$$\begin{aligned} \psi(t) &\leq \alpha(t) - \alpha(t) + \alpha(0)B(t) + \int_0^t \alpha'(s)\frac{B(t)}{B(s)}ds \\ &= \alpha(0)B(t) + \int_0^t \alpha'(s)\frac{B(t)}{B(s)}ds, \end{aligned}$$

which is the desired alternative form of Gronwall's inequality.

Integrating from 0 to t and using $A(0) = 0$, $B(0) = 1$, we obtain

$$\begin{aligned} A(t)B^{-1}(t) &\leq \int_0^t \alpha(s)\beta(s)B^{-1}(s)ds, \\ A(t) &\leq \int_0^t \alpha(s)\beta(s)\frac{B(t)}{B(s)}ds, \end{aligned}$$

where in the last line we multiplied by $B(t) > 0$, so the inequality is preserved.

Using this result with the first assumption (20), we get that

$$\psi(t) \leq \alpha(t) + A(t) \leq \alpha(t) + \int_0^t \alpha(s)\beta(s)\frac{B(t)}{B(s)}ds.$$

□

9.2 Perturbation Theory

Suppose we have the following ODEs and initial conditions, where y_0 and g are perturbations of x_0 and f :

$$\begin{aligned} \frac{d}{dt}x &= f(t, x), & \frac{d}{dt}y &= g(t, y) \\ x(t_0) &= x_0, & y(t_0) &= y_0. \end{aligned}$$

Suppose there is a small perturbation (or measurement error) in x_0 or f . We wonder: How do these errors affect x ? We hope that the change in x is small.

Theorem 9.2.1. (Continuous dependence in initial condition and parameters) Suppose f is Lipschitz and the difference between x and y is bounded

$$|f(t, y_1) - f(t, y_2)| \leq L |y_1 - y_2| \quad (22)$$

$$\max_{(t,x) \in D} |f(t, x) - g(t, x)| = M. \quad (23)$$

Then

$$|x(t) - y(t)| \leq e^{L(t-t_0)} |x_0 - y_0| + \frac{M}{L} [e^{L(t-t_0)} - 1].$$

Corollary 9.2.2. (i) Suppose $f = g$ and $x_0 = y_0$. Then $x(t) = y(t)$, so we know the solution is unique!

(ii) Suppose $f = g$ and define $\phi(t, x_0)$ where

$$\frac{d}{dt}\Phi(t) = f(t, \Phi), \quad \Phi(t_0, x_0) = x_0.$$

Since

$$|x(t) - y(t)| \leq e^{L(t-t_0)} |x_0 - y_0|$$

because $M = 0$, we get that the mapping $x_0 \mapsto \Phi(t, x_0)$ is Lipschitz in x_0 .

Φ gives solutions to the ODE for different initial conditions.

(iii) Suppose $x_0 = y_0$, $g(t) = f(t) + o(\epsilon)$, where $|f - g| \leq \epsilon$. Then

$$|x(t) - y(t)| \leq \frac{\epsilon}{L} \left(e^{L(t-t_0)} - 1 \right).$$

Proof. We start with

$$\begin{aligned} \frac{d}{dt} [x(t) - y(t)] &= f(t, x(t)) - g(t, x(t)) \\ x(t) - y(t) &= x_0 - y_0 + \int_0^t f(s, x(s)) - g(s, y(s)) ds \\ |x(t) - y(t)| &\leq |x_0 - y_0| + \int_0^t |f(s, x(s)) - g(s, y(s))| ds \end{aligned} \quad (24)$$

Noting that

$$|f(s, x(s)) - g(s, y(s))| \leq |f(s, x(s)) - f(s, y(s))| + |f(s, y(s)) - g(s, y(s))|,$$

we see that the first assumption (22) implies the first term is less than $L|x(s) - y(s)|$ and the second assumption (23) implies that the second term is less than M . Hence

$$|f(s, x(s)) - g(s, y(s))| \leq L|x(s) - y(s)| + M,$$

so we substitute this inequality into (24) to get

$$|x(t) - y(t)| \leq |x_0 - y_0| + \int_{t_0}^t (L|x(s) - y(s)| + M) ds.$$

Define

$$\psi(t) := |x(t) - y(t)|, \quad \alpha(t) := |x_0 - y_0| + M(t - t_0), \quad \beta(t) := L.$$

Then

$$\psi(t) \leq \alpha(t) + \int_{t_0}^t \beta(s) \psi(s) ds$$

and we can apply Gronwall's inequality in the form

$$\psi(t) \leq \alpha(t_0) \exp \left(\int_{t_0}^t \beta(u) du \right) + \int_{t_0}^t \alpha'(s) \exp \left(\int_s^t \beta(u) du \right) ds.$$

Since $\alpha(t_0) = |x_0 - y_0|$, $\alpha'(s) = M$, and $\beta(u) = L$, this gives

$$\begin{aligned} |x(t) - y(t)| &= \psi(t) \leq |x_0 - y_0| e^{L(t-t_0)} + M \int_{t_0}^t e^{L(t-s)} ds \\ &= |x_0 - y_0| e^{L(t-t_0)} + \frac{M}{L} (e^{L(t-t_0)} - 1), \end{aligned}$$

which is the desired estimate. $\square$

Example 9.2.3. Suppose we have an ODE we cannot solve by hand:

$$\frac{d}{dt} x = \frac{x^2}{1 + t^2 + x^2}, \quad x(0) = 0.$$

If there is a small perturbation in x to y , where

$$\frac{d}{dt}y = \frac{y^2 + \epsilon}{1 + t^2 + y^2}, \quad y(0) = \epsilon,$$

because $|f - g| \leq \epsilon$ and $|x_0 - y_0| \leq \epsilon$, and f is Lipschitz in x , we know that

$$|x(t) - y(t)| \leq \epsilon e^{Lt} + \frac{\epsilon}{L} (e^{Lt} - 1).$$

SUPPOSE WE KNOW that

$$y' \leq F(t, y(t)), \quad t \in [a, b]$$

for some F that makes the equation unsolvable. What can we say about ???

Theorem 9.2.4. (Comparison) Suppose F is Lipschitz and f, g are differentiable. If

$$f' \leq F(t, f(t)), \quad \forall t \in [a, b] \quad (25)$$

$$g \text{ solves } g' = F(t, g(t)) \quad (26)$$

$$f(a) \leq g(a). \quad (27)$$

Then

$$f(t) \leq g(t), \quad \forall t \in [a, b].$$

Proof. Suppose for contradiction that there exists $t_1 \in [a, b]$ such that $f(t_1) > g(t_1)$, where $t_1 > a$. Define

$$\Omega = \{t : f(t) \leq g(t)\}.$$

Note that $\Omega \neq \emptyset$ because $a \in \Omega$. Let

$$t_0 = \sup\{t \in [a, t_1] \cap \Omega\}.$$

(Step 1) We first record what the choice of t_0 implies. By definition of t_0 as a supremum, we have:

- (i) $f(t) \leq g(t)$ for all $t \in [a, t_0]$,
- (ii) $f(t) > g(t)$ for all $t \in (t_0, t_1]$.

Since $t_0 = \sup([a, t_1] \cap \Omega)$, there exists a sequence $t_n \in [a, t_1] \cap \Omega$ with $t_n \rightarrow t_0$. For each n , $f(t_n) \leq g(t_n)$, so by continuity of f and g ,

$$f(t_0) \leq g(t_0).$$

On the other hand, by (ii) we have $f(t) > g(t)$ for $t > t_0$ close to t_0 , so continuity again gives

$$f(t_0) \geq g(t_0).$$

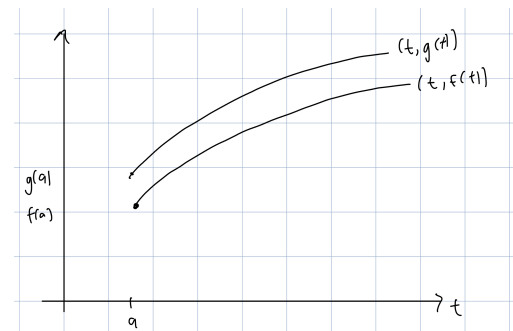


Figure 1: f must be bounded by the known g .

Hence

$$f(t_0) = g(t_0).$$

(Step 2) By the first and second assumptions (25, 26), we get that

$$\begin{aligned} f'(t) - g'(t) &\leq F(t, f(t)) - F(t, g(t)) \\ \implies f(t) - g(t) &\leq f(t_0) - g(t_0) + \int_{t_0}^t F(s, f(s)) - F(s, g(s)) ds \\ f(t) - g(t) &\leq \underbrace{f(t_0) - g(t_0)}_{=0} + \int_{t_0}^t F(s, f(s)) - F(s, g(s)) ds \\ &\leq \int_{t_0}^t L |f(s) - g(s)| ds \\ &\leq \int_{t_0}^t L(f(s) - g(s)) ds \end{aligned}$$

for every $t \in [t_0, t_1]$, since at these values, $f(t) \geq g(t)$. Gronwall's inequality gives us $f(t) - g(t) \leq 0$ for every $t \in [t_0, t_1]$, which contradicts $f(t_1) - g(t_1) > 0$. $\square$

10 Thursday, October 30: Comparison, Extension, and Blowup

10.1 Comparison

Suppose we know that

$$y'(t) \leq F(t, y(t)), t \in [a, b] \quad (28)$$

We prove the following theorem last class.

Theorem 10.1.1. Suppose that F is Lipschitz in y such that

$$f'(t) \leq F(t, f(t)) \quad (29)$$

$$g'(t) = F(t, g(t)) \quad (30)$$

$$\exists f(a) \leq g(a). \quad (31)$$

then $f(t) \leq g(t)$ for all $t \in [a, b]$.

Example 10.1.2. What happens when we have there is no satisfactory Lipschitz F ? Suppose

$$\begin{aligned} \frac{d}{dt}y &= y^{\frac{1}{3}} \\ y(0) &= 0 \\ g &= 0. \end{aligned}$$

Then $y(t) = ct^\alpha$ (?) so

$$\frac{d}{dt}y = c\alpha t^{\alpha-1} = \frac{c_1}{3} t^{\frac{\alpha}{3}}.$$

Center $\alpha = \frac{3}{2}$ and $c^{\frac{2}{3}} = \frac{1}{\alpha}$. We get that $C_{\pm} = \pm(\frac{1}{\alpha})^{\frac{3}{2}}$. Then $|t|^{\frac{3}{2}}$ is a C^1 solution to the ODE, but f is then not always less than g .

Theorem 10.1.3. Suppose

$$\frac{d}{dt}x = f(t, x), \quad \frac{d}{dt}y = g(t, y).$$

and we have solutions x, y in the interval $t \in [a, b]$. With the assumptions

- (i) $f(t, z) \leq g(t, z), \forall t \in [a, b], z \in D$
- (ii) f or g is locally Lipschitz in z on the region given by $t \in [a, b], z \in D$
- (iii) $x(a) \leq y(a)$.

Then $x(t) \leq y(t)$ for all $t \in [a, b]$

Proof. We just prove this for if g is Lipschitz. Then

$$\begin{aligned} \frac{d}{dt}x(t) &= f(t, x) < g(t, x) \\ \frac{d}{dt}y &= g(t, y). \end{aligned}$$

We can use the previous theorem. □

Example 10.1.4. Suppose we have a family of ODEs

$$\frac{d}{dt}x_{\alpha}(t) = f(t, x_{\alpha}(t)) + \alpha$$

with the initial conditions

$$x_{\alpha}(t_0) = x_0.$$

Then the functions are increasing in α .

Example 10.1.5. Suppose $\frac{d}{dt}x(t) = P_n(x, t)$ for a polynomial P_n . Since $P_n \leq ce^x$ eventually, we can use $\frac{d}{dt}y(t) = ce^y$ and $y(0) = x_0$ to see that for wherever we can solve y , eventually $x(t) \leq y(t)$.

10.2 Extensibility of Solutions

Recall the existence-uniqueness theorem:

Theorem 10.2.1. Given $\frac{d}{dt}x = f(t, x)$ and $x(t_0) = x_0$, some rectangle defined by $|t - t_0| < a, |x - x_0| < b$ on which f is Lipschitz in x , and that f is bounded by M on the rectangle, there exists a unique C^1 solution for $t \in [t_0 - h, t_0 + h]$ for

$$h < \min\left(\frac{1}{L}, \frac{b}{M}, a\right).$$

This theorem only guarantees a solution in a small neighborhood of (t_0, x_0) .

Example 10.2.2. (Blowup) Consider $\frac{d}{dt}y = y^2$ and $y(0) = 1$. The solution is given by

$$y(t) = \frac{1}{1-t}$$

in a neighborhood. However,

$$\lim_{t \rightarrow 1^-} y(t) = \infty,$$

so we say this solution exists up to 1^- .

How can we characterize (t^-, t^+) where the solution exists?

Lemma 10.2.3. (Gluing Lemma) Suppose $f(t, x(t)) \in C(\mathbb{R} \times \mathbb{R})$ and Lipschitz with respect to a function L_D in x in any bounded region D such that L_D is bounded on D . Suppose $\phi_i(t)$ solves

$$\frac{d}{dt}\phi_i(t) = f(t, \phi(t)), \phi(t_0) = x_0$$

when $t \in I_i(a_i, b_i)$ for $i \in \{1, 2\}$.

If $t_0 \in I_1 \cap I_2$ and

$$\phi_1(t_0) = \phi_2(t_0),$$

then

$$\phi_1(t) = \phi_2(t)$$

for $t \in I_1 \cap I_2$ and

$$\phi(t) = \begin{cases} \phi_1(t), & t \in I_1 \\ \phi_2(t), & t \in I_2 \end{cases}$$

is a C^1 solution to the ODE on $I_1 \cup I_2$.

Proof. Apply the existence-uniqueness theorem to

$$\frac{d}{dt}x(t) = f(t, x(t)), \quad x(t_0) = x_0.$$

There exists a unique C^1 solution $x(t)$ on some interval $[t_0 - h, t_0 + h]$.

By uniqueness, $\phi_1(t) = \phi_2(t)$ for $t \in [t_0 - h, t_0 + h] =: [a, b]$.

Define

$$J^- = \{p : \phi_1(t) = \phi_2(t), t \in [p, t_0 + h] \cap (a, b)\}.$$

Define

$$A = \inf J^-.$$

We want to show that $A = a$.

(Method 1) Define $J = \{t \in I_1 \cap I_2 : \phi_1(t) = \phi_2(t)\}$. J is relative closed (by continuity) and relative open (by step 1). It is nonempty, so $J = I_1 \cap I_2$.

(Method 2) By definition of infimum, there exists a sequence $\{p_n\} \subset J^-$ that converges to A . Suppose for contradiction that $A > a$. By definition of J^- , for each $n \in \mathbb{N}$,

$$\phi_1(p_n) = \phi_2(p_n).$$

Sending $n \rightarrow \infty$,

$$\phi_1(A) = \phi_2(A),$$

so $A = a$. Using existence-uniqueness, we can replace (t_0, x_0) with $(A, \phi_1(A))$ so $\phi_1(t) = \phi_2(t)$ for $t \in [A - h, A + h]$, contradicting A being the infimum of J^- . The two solutions must agree at the end endpoint (and we can reproduce it with sup to extend to b).

Now we know that ϕ_1 and ϕ_2 agree on $[a, b]$. ϕ as given below is well-defined:

$$\phi(t) = \begin{cases} \phi_1(t), & t \in I_1 \\ \phi_2(t), & t \in I_2 \end{cases}.$$

ϕ is $C^1(I_1 \cup I_2)$. It is also a solution to the ODE $\phi' = f(t, \phi)$, as

$$\begin{cases} \phi'(t) = \phi_1'(t) = f(t, \phi_1) = f(t, \phi), & t \in I_1 \\ \phi'(t) = \phi_2'(t) = f(t, \phi_2) = f(t, \phi), & t \in I_2 \end{cases}.$$

□

11 Thursday, November 6: Extension and Blowup

11.1 Review

Consider

$$\frac{d}{dt}x = f(t, x(t)),$$

where $f \in C(R \times R)$ and f is Lipschitz in its second argument in any bounded domain D . These assumptions will hold for the entirety of this lecture.

Recall the gluing lemma:

Lemma 11.1.1. Suppose that $\phi_i(t)$ solves the ODE, $I_i = (a_i, b_i)$, and $t_0 \in I_1 \cap I_2 = (a, b) \neq \emptyset$.

If $\phi_1(t_0) = \phi_2(t_0)$, then

$$\phi_1(t) = \phi_2(t)$$

for $t \in I_1 \cap I_2$, so we can define

$$\phi(t) = \begin{cases} \phi_1(t), & t \in I_1, \\ \phi_2(t), & t \in I_2 \end{cases}$$

for $t \in I_1 \cup I_2$, where ϕ is the unique C^1 solution to the ODE.

Note that this is automatically implied if $\frac{\partial f}{\partial y}$ exists.

An alternative proof uses Gronwall's inequality with

$$|\phi_1(t) - \phi_2(t)| \leq \int_{t_0}^t L_D |\phi_1(s) - \phi_2(s)| ds$$

for $t \in I_1 \cap I_2$ to prove the first, trickier part of the lemma result.

Now we wonder what happens if f is not Lipschitz but we have two solutions in overlapping domains. By drawing a picture, the first conclusion is not true, while ϕ in the second conclusion is not well-defined. There are multiple choices for how to glue ϕ_1 and ϕ_2 together.

11.2 Blowup

Theorem 11.2.1. (*Blowup Criterion for Extension*) Suppose that

$$\frac{d}{dt}\phi(t) = f(t, \phi(t))$$

for $t \in (t_-, t_+)$, where $f \in C(\mathbb{R} \times \mathbb{R})$ and f is Lipschitz in its second argument. The solution can extend to $(t_-, t_+ + \epsilon)$ for some $\epsilon > 0$ if and only if there exists a sequence $\{t_n\}$ such that

$$t_n \rightarrow (t_+)^-, \quad \phi(t_n) \rightarrow \phi_0 \in \mathbb{R}.$$

We can also write this as

$$(t_n, \phi(t_n)) \rightarrow (t_+, \phi_0).$$

Alternatively, the solution extends if and only if

$$\liminf_{t \rightarrow (t_+)^-} |\phi(t)| < \infty.$$

We can reformulate the theorem to get

Theorem 11.2.2. (*Blowup Criterion for Extension, Alternative*) $\phi(t)$ cannot extend beyond t_+ if and only if

$$\lim_{t \rightarrow (t_+)^-} |\phi(t)| = \infty.$$

Proof. (Of Theorem (11.2.2)) Since ϕ is continuous, extensibility implies that the limit as $t \rightarrow t_+$ from the left exists and is finite (in fact, equal to $\phi(t_+)$, so the forwards direction is immediate.

For the backwards direction, suppose there exists a sequence $\{t_n\}$

$$(t_n, \phi(t_n)) \rightarrow (t_+, \phi_0).$$

Then there exists $N \in \mathbb{N}$ such that for $n \geq N$,

$$|\phi(t_n) - \phi_0| \leq 1.$$

We want to use existence-uniqueness to get a $\phi_{\text{new}}(t)$ that solves the ODE on some open interval that contains t_+ .

(Step 1) First define

$$D = [t_+ - 2, t_+ + 2] \times [\phi_0 - 2, \phi_0 + 2].$$

By the two assumptions,

$$|f| \leq M \text{ in } D, \quad |f(t, x) - f(t, y)| \leq L|x - y|.$$

Importantly, M and L are independent of n ; they only depend on the accumulation point.

(Step 2) We apply existence-uniqueness with the initial condition $(t_n, \phi(t_n))$ to get a solution

$$\phi_{new}(t_n) = \phi(t_n)$$

for $t \in (t_n - h, t_n + h)$, where

$$h < \frac{1}{2} \min\{1, \frac{1}{M}, \frac{1}{L}\}.$$

Again, we see that h is independent of n — in particular, as $n \rightarrow \infty$, h remains positively bounded above 0.

(Step 3) Hence, we can pick n large enough such that

$$t_n + h > t_+.$$

We have two solutions

$$\begin{cases} \phi_{new}(t), & t \in (t_n - h, t_n + h) \\ \phi(t), & t \in (t_-, t_+) \end{cases}$$

that both solve the ODE and coincide with $\phi(t_n) = \phi_{new}(t_n)$, so we can use the gluing lemma to extend ϕ to be defined on

$$(t_-, t_n + h) \subseteq (t_-, t_+) \cup (t_n - h, t_n + h),$$

where $t_+ + \epsilon$ is contained in the interval. $\square$

A crucial part of the proof was noting that f is bounded and Lipschitz in some large region, then solving the ODE on a smaller subregion so ϕ_{new} is within the large region.

why?

A priori, we could not exclude an oscillation near the limiting point, but after completing the proof, we know that the solution ϕ is C^1 .

how can ϕ wildly oscillate if $\phi' = f$ is bounded by M ???

11.3 Global Solutions

What kinds of ODEs have global solutions? Can we think of any that we cannot solve by hand? This question may seem ill-posed, but we can actually answer it using the blowup criterion.

Suppose

$$\frac{d}{dt}x = f(t, x),$$

where $|f| < M$; i.e., f is uniformly bounded. We claim that this ODE has a unique (up to initial condition (?)) global C^1 solution.

Supposing not, in that we only have a solution on (t_-, t_+) , we get

$$\begin{aligned} |x(t) - x(t_0)| &= \left| \int_{t_0}^t f(s, x(s)) ds \right| \\ &\leq M |t - t_0| < \infty. \end{aligned}$$

We get that any sequence $\{t_n\} \rightarrow t_+$ has a subsequence such that $\{x(t_n)\}$ converges, so x extends by the blowup criterion.

11.4 Finite-time Blowup

Finite-time blowup means that the derivative of the solution goes to infinity near some point; it is the opposite of existence of a global solution.

Suppose we have

$$\frac{d}{dt}y = y^p, \quad y(0) = 1.$$

When $p = 1$, $y(t) = e^t$.

Consider when $p > 1$. Because $y(0) > 1$, we know that $y^p \geq y$. Using separation of variables,

$$\begin{aligned} \frac{\frac{d}{dt}y}{y^p} &= 1 \\ \implies y^{1-p}(t) &= 1 + (1-p)t \\ y^{p-1}(t) &= \frac{1}{\underbrace{1 - (p-1)t}_{>0}} \end{aligned}$$

so $y^{p-1}(t) \rightarrow \infty$ as $t \rightarrow \frac{1}{p-1}$. We see that with faster-than-exponential growth, the solution goes to ∞ in finite time, so $p = 1$ is a borderline case.

Recall from comparison that if

$$\frac{d}{dt}x = f(t, x), \quad \frac{d}{dt}y = g(t, y),$$

where

$$f \leq g, \quad x(0) < y(0),$$

we know that

$$x(t) \leq y(t)$$

for $t \in I_x \cap I_y$.

Example 11.4.1. Consider

$$\frac{d}{dt}y = e^y.$$

By comparison ($e^y \geq y^2$), we see that there does not exist a global solution. The same thing applies to the ODE

$$\frac{d}{dt}y = y^n + \dots + 1 + e^t$$

for $n \geq 2$.

Example 11.4.2. Consider

$$\frac{d}{dt}y = y^2, \quad y(0) = y_0.$$

If $y_0 = 0$, then $y = 0$. If $y_0 > 0$, then we see blowup as $t \rightarrow \infty$. If $y_0 < 0$, we get the solution formula

$$y(t) = \frac{y_0}{1 - ty_0} \sim -\frac{1}{t}.$$

So, y starts below 0 then increases to 0^- as $t \rightarrow \infty$.

This ODE is interesting; a small perturbation from y_0 in the positive direction leads to a large change in the solution (we now have blowup), but a perturbation in the negative direction leads to equivalent behavior in the distant horizon.

Example 11.4.3. What if we now perturb with

$$\frac{d}{dt}y = y^2 + \epsilon, \quad y(0) = y_0?$$

Since

$$y(t) \geq y_0 + \epsilon t,$$

which is eventually positive, we see that $y(t_0) > 0$ for t_0 large. Hence

$$\frac{d}{dt}y \geq y^2,$$

so we have blowup.

Example 11.4.4. If we have

$$\frac{d}{dt}y = y^2 - \epsilon^2, \quad y_0 \in (-\epsilon, \epsilon),$$

we get very different behaviors around $-\epsilon, \epsilon$.

12 Blowup Criterion, Redone

12.1 Blowup

Definition 12.1.1 (Maximal solution). Let $f: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be continuous and locally Lipschitz in its second argument. A solution $\phi: I \rightarrow \mathbb{R}$ of

$$\phi'(t) = f(t, \phi(t))$$

on an interval $I \subset \mathbb{R}$ is called *maximal* if there is no strictly larger interval $J \supsetneq I$ and solution $\tilde{\phi}: J \rightarrow \mathbb{R}$ such that $\tilde{\phi}(t) = \phi(t)$ for all $t \in I$.

Theorem 12.1.2 (Blowup criterion). *Consider the IVP*

$$\phi'(t) = f(t, \phi(t)), \quad t \in (t_-, t_+),$$

where $f \in C(\mathbb{R} \times \mathbb{R})$ and f is locally Lipschitz in its second argument. Let $\phi: (t_-, t_+) \rightarrow \mathbb{R}$ be a solution.

Then the following are equivalent:

(i) ϕ extends to a solution on a larger interval $(t_-, t_+ + \varepsilon)$ for some $\varepsilon > 0$;

(ii) there exists a sequence $t_n \in (t_-, t_+)$ with

$$t_n \rightarrow t_+^-, \quad \phi(t_n) \rightarrow \phi_0 \in \mathbb{R},$$

i.e. the graph $(t_n, \phi(t_n))$ has a finite limit point (t_+, ϕ_0) ;

(iii)

$$\liminf_{t \rightarrow t_+^-} |\phi(t)| < \infty.$$

Remark 12.1.3. Geometrically, we are looking at the curve $t \mapsto (t, \phi(t))$ in the (t, x) -plane. The theorem says: the solution can be continued a little past t_+ if and only if, as $t \rightarrow t_+^-$, the curve stays inside some vertical strip $\{(t, x) : t < t_+, |x| \leq R\}$. The only obstruction to continuation is that $|\phi(t)|$ blows up in finite time.

If the solution does not “run off to infinity” in finite time, then its graph accumulates at some finite point (t_+, ϕ_0) , and from that point we can restart the IVP and continue the solution a bit further in t .

Proof. We will prove (i) $\Leftrightarrow$ (ii), and then see that (ii) $\Leftrightarrow$ (iii) is just a basic fact about $\liminf$.

(i) $\Rightarrow$ (ii). Suppose ϕ extends to a solution on $(t_-, t_+ + \varepsilon)$ for some $\varepsilon > 0$. Denote this extension by $\tilde{\phi}: (t_-, t_+ + \varepsilon) \rightarrow \mathbb{R}$, so $\tilde{\phi}(t) = \phi(t)$ for $t \in (t_-, t_+)$ and $\tilde{\phi}$ satisfies

$$\tilde{\phi}'(t) = f(t, \tilde{\phi}(t))$$

for all $t \in (t_-, t_+ + \varepsilon)$. Since $\tilde{\phi}$ is continuous, the one-sided limit

$$\tilde{\phi}(t_+) = \lim_{t \rightarrow t_+} \tilde{\phi}(t)$$

exists and is finite.

Now fix any sequence $t_n \in (t_-, t_+)$ with $t_n \rightarrow t_+^-$. Then

$$\phi(t_n) = \tilde{\phi}(t_n) \rightarrow \tilde{\phi}(t_+) = \phi_0,$$

which is exactly condition (ii).

(ii) $\Rightarrow$ (iii). If we have $t_n \rightarrow t_+^-$ and $\phi(t_n) \rightarrow \phi_0 \in \mathbb{R}$, then the sequence $\{\phi(t_n)\}$ is bounded. By definition of the liminf,

$$\liminf_{t \rightarrow t_+^-} |\phi(t)| \leq \lim_{n \rightarrow \infty} |\phi(t_n)| = |\phi_0| < \infty,$$

so (iii) holds.

(iii) $\Rightarrow$ (ii). This is a purely real-analysis statement: for any real-valued function g ,

$$\liminf_{t \rightarrow t_+^-} |g(t)| < \infty \iff \exists t_n \rightarrow t_+^- \text{ with } |g(t_n)| \text{ bounded.}$$

Applying the Bolzano–Weierstrass theorem to the bounded sequence $\{g(t_n)\}$, we can pass to a subsequence (which we still call t_n) such that $g(t_n)$ converges. Applied to $g = \phi$, this gives exactly (ii).

(ii) $\Rightarrow$ (i). This is the nontrivial direction: we show that if the solution has a finite limit point at t_+ , then it can be extended beyond t_+ .

Assume there exists a sequence $t_n \rightarrow t_+^-$ such that

$$\phi(t_n) \rightarrow \phi_0 \in \mathbb{R}.$$

Consider the point (t_+, ϕ_0) in the (t, x) -plane. Since f is continuous and locally Lipschitz in x , there is a closed rectangle

$$Q : [t_+ - a, t_+ + a] \times [\phi_0 - b, \phi_0 + b]$$

with $a, b > 0$ such that:

- (1) f is bounded on Q , say $|f(t, x)| \leq M$ for all $(t, x) \in Q$;
- (2) f is Lipschitz in x on Q with some constant L .

By the usual proof of the Picard–Lindelöf theorem, there exists $h > 0$ (depending only on a, b, M) such that for every $(s, x) \in Q$ there is a unique solution $y(\cdot; s, x)$ of

$$y'(t) = f(t, y(t)), \quad y(s) = x,$$

defined at least on the interval $[s - h, s + h] \subset (t_+ - a, t_+ + a)$.

In particular:

- there is a unique solution ψ with

$$\psi'(t) = f(t, \psi(t)), \quad \psi(t_+) = \phi_0,$$

defined on $(t_+ - h, t_+ + h)$;

- for all n large enough, $(t_n, \phi(t_n)) \in Q$, so there is a unique solution ψ_n with

$$\psi_n'(t) = f(t, \psi_n(t)), \quad \psi_n(t_n) = \phi(t_n),$$

defined on $(t_n - h, t_n + h)$.

We are using here the quantitative version of the existence–uniqueness theorem: on any compact rectangle where f and its Lipschitz constant are bounded, we get a uniform existence time $h > 0$ for all initial values in a slightly smaller rectangle.

Now note:

- Since ϕ is already a solution with $\phi(t_n) = \psi_n(t_n)$, uniqueness gives

$$\psi_n(t) = \phi(t) \quad \text{for all } t \in (t_-, t_+) \cap (t_n - h, t_n + h).$$

In particular, for n large enough we have

$$\psi_n(t) = \phi(t) \quad \text{for all } t \in (t_+ - h/2, t_+).$$

- The continuous dependence of solutions on initial data (proved earlier via Gronwall) says that if $(t_n, \phi(t_n)) \rightarrow (t_+, \phi_0)$, then

$$\psi_n \rightarrow \psi \quad \text{uniformly on any compact subinterval of } (t_+ - h, t_+ + h).$$

Fix a compact interval $[t_0, t_+] \subset (t_+ - h, t_+)$. For n large enough, we have $\psi_n(t) = \phi(t)$ on $[t_0, t_+]$, and $\psi_n \rightarrow \psi$ uniformly on $[t_0, t_+]$. Hence

$$\psi(t) = \lim_{n \rightarrow \infty} \psi_n(t) = \lim_{n \rightarrow \infty} \phi(t) = \phi(t) \quad \text{for all } t \in [t_0, t_+].$$

Since t_0 was arbitrary in $(t_+ - h, t_+)$, this shows

$$\psi(t) = \phi(t) \quad \text{for all } t \in (t_+ - h, t_+).$$

Thus ψ is an honest extension of ϕ beyond t_+ : define

$$\tilde{\phi}(t) = \begin{cases} \phi(t), & t \in (t_-, t_+), \\ \psi(t), & t \in [t_+, t_+ + h]. \end{cases}$$

Then $\tilde{\phi}$ is C^1 on $(t_-, t_+ + h)$, satisfies $\tilde{\phi}'(t) = f(t, \tilde{\phi}(t))$ for all $t \in (t_-, t_+ + h)$, and extends ϕ to the larger interval $(t_-, t_+ + h)$. This is exactly (i).

We have now shown (i) $\Leftrightarrow$ (ii) and (ii) $\Leftrightarrow$ (iii), so all three statements are equivalent. $\square$

Theorem 12.1.4 (Blowup criterion, alternative form). *Let $\phi: (t_-, t_+) \rightarrow \mathbb{R}$ be a maximal solution of*

$$\phi'(t) = f(t, \phi(t)),$$

with f as above, and suppose $t_+ < \infty$. Then ϕ cannot be extended beyond t_+ if and only if

$$\lim_{t \rightarrow t_+^-} |\phi(t)| = \infty.$$

In other words, finite-time breakdown of a maximal solution can only occur by blowup of the solution itself.

Proof. Since ϕ is maximal and $t_+ < \infty$, by definition it cannot be extended past t_+ . By Theorem 12.1.2, this means that

$$\liminf_{t \rightarrow t_+} |\phi(t)| = \infty.$$

But $\liminf_{t \rightarrow t_+} |\phi(t)| = \infty$ is equivalent to

$$\forall M > 0 \exists \delta > 0 \text{ such that } t_+ - \delta < t < t_+ \implies |\phi(t)| > M,$$

which is exactly the statement that $|\phi(t)| \rightarrow \infty$ as $t \rightarrow t_+$, i.e.

$$\lim_{t \rightarrow t_+} |\phi(t)| = \infty.$$

Conversely, if $|\phi(t)| \rightarrow \infty$ as $t \rightarrow t_+$ but ϕ could be extended to a larger interval $(t_-, t_+ + \epsilon)$, the extension would be continuous on a compact interval $[t_+ - \epsilon/2, t_+]$ and hence bounded there, contradicting $|\phi(t)| \rightarrow \infty$. $\square$

Remark 12.1.5. *There is a completely analogous statement at the left endpoint t_- : if the maximal solution has $t_- > -\infty$, then ϕ cannot be extended to the left if and only if $|\phi(t)| \rightarrow \infty$ as $t \rightarrow t_-$.*

13 Autonomous ODEs, Stability

13.1 Preliminaries

Definition 13.1.1 (Autonomous ODE). An autonomous ODE is one we can write in the form

$$\frac{d}{dt} \vec{x} = F(\vec{x}).$$

In other words, $\frac{d}{dt} \vec{x}$ does not depend on t .

With an autonomous ODE, if we find a point x^* at which $F(x^*) = 0$, we get that the unique solution to the ODE is a constant. We refer to x^* as a critical point.

Definition 13.1.2 (Trajectory). The trajectory of an ODE is the curve defined by $(t, X(t))$.

When $X(t) \in \mathbb{R}^2$, we can think of x_2 as a function of x_1 ⁴ by the implicit function theorem. Varying t , we can plot x_2 against x_1 to get a graph in the *phase plane*: the $x_1 x_2$ - plane.

⁴ As long as $X' = (x'_1, x'_2) \neq 0$

Definition 13.1.3 (Phase portrait). The representative set of a trajectory is called the phase portrait. By representative set, we mean the set of all trajectories attained by varying $x_{1,0}$ and $x_{2,0}$.

Example 13.1.4. Consider the autonomous ODE $\frac{d}{dt} x = x^2 - x$. We want to set $f = x^2 - x = 0$, so our critical points are

$$x = 0, \quad x = 1.$$

Then we have two special solutions $x_1(t) = 0$ and $x_2(t) = 1$.

Example 13.1.5. Now consider the autonomous ODE system

$$\begin{cases} \frac{d}{dt}x = x^2 - 2y \\ \frac{d}{dt}y = x + y \end{cases}.$$

Setting them equal to 0, we get the system

$$x^2 = 2y, \quad x = -y.$$

Hence our two critical points are

$$(x, y) = (0, 0), \quad (-2, 2).$$

13.2 Linear 2×2 System

In this section, consider $\frac{d}{dt}X = AX$, where $A \in \mathbb{R}^{2 \times 2}$.

Note that we have an equilibrium for X being the eigenvector associated with an eigenvalue of 0; i.e., the kernel of A .

In general, if $Av_i = \lambda_i v_i$, we have the solution

$$x(t) = c_1 e^{\lambda_1 t} v_1 + c_2 e^{\lambda_2 t} v_2.$$

Recall the three cases we have:

- (I) $\lambda_1 < \lambda_2$ and are both real,
- (II) $\lambda_1 = \overline{\lambda_2}$, and
- (III) $\lambda_1 = \lambda_2$ is real.

IN CASE (I), we can write the general solution as

$$x(t) = e^{\lambda_2 t} \left(e^{(\lambda_1 - \lambda_2)t} c_1 v_1 + c_2 v_2 \right).$$

When $\lambda_1 < \lambda_2 < 0$, we see that if either $c_1 = 0$ or $c_2 = 0$, then x goes to 0 as time goes to ∞ :

$$x_1(t) = c_1 e^{\lambda_1 t} v_1 \xrightarrow{t \rightarrow \infty} 0.$$

Similarly,

$$x_2(t) = c_2 e^{\lambda_2 t} v_2 \xrightarrow{t \rightarrow \infty} 0$$

by putting $c_1 = 0$. If $c_1, c_2 \neq 0$, then we look at the general solution and see that since $\lambda_1 < \lambda_2$, the first term inside the parentheses goes to 0. Hence the asymptotic behavior is like $e^{\lambda_2 t} (c_2 v_2)$, which goes to 0.

When $\lambda_1 < 0 < \lambda_2$, we see that $x_1 \rightarrow 0$ but x_2 goes to either ∞ or $-\infty$, depending on the sign of c_2 . The asymptotic behavior of x has not changed: $x(t) \approx e^{\lambda_2 t} c_2 v_2$. Hence, $x(t) \rightarrow 0$ iff $c_2 = 0$.

When $0 < \lambda_1 < \lambda_2$, both x_1 and x_2 go to ∞ or $-\infty$. Again, the asymptotic behavior is the same, so $x(t) \rightarrow 0$ iff $c_1 = 0$.

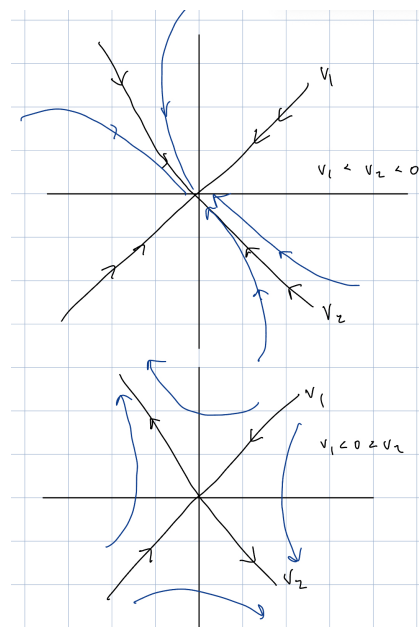


Figure 2: Case (I)

IN CASE (II), write

$$\begin{aligned}\lambda_{\pm} &= \alpha \pm i\beta \\ v_{\pm} &= a \pm ib.\end{aligned}$$

Then

$$\begin{aligned}x(t) &= c_1 \operatorname{Re} \left(e^{\lambda_+ t} v_+ \right) + c_2 \operatorname{Im} \left(e^{\lambda_+ t} v_+ \right) \\ &= e^{\alpha t} [c_1 (\cos(\beta t)a - \sin(\beta t)b) + c_2 (\sin(\beta t)a + \cos(\beta t)b)] \\ &= e^{\alpha t} v(t).\end{aligned}$$

We see that $|x|$ has the behavior

$$\begin{cases} |x(t)| \rightarrow \infty, & \alpha > 0 \\ |x(t)| = 1, & \alpha = 0 \\ |x(t)| \rightarrow 0, & \alpha < 0 \end{cases}.$$

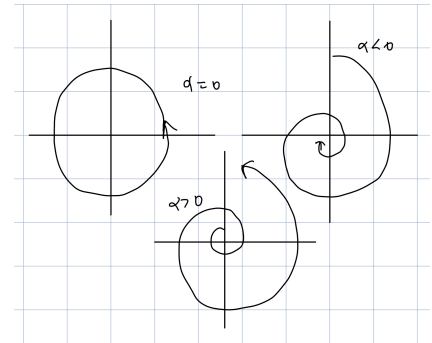


Figure 3: Case (II)

IN CASE (III), we have $\lambda_1 = \lambda_2 = \lambda$. When we have two distinct eigenvectors v_1, v_2 , we can write

$$x(t) = e^{\lambda t} \underbrace{(c_1 v_1 + c_2 v_2)}_{\text{constant}}.$$

Hence, the trajectory is affine and depending on if $\lambda < 0$, $\lambda = 0$, or $\lambda > 0$, we see that we either $(0,0)$ is a source, sink, or the trajectory is just constant at the initial point.

When we have one eigenvector and one generalized eigenvector, we have to write

$$x(t) = e^{\lambda t} (c_1 v_1 + c_2 t v_2),$$

and the term in the parentheses is no longer constant. But as we send $t \rightarrow \infty$, eventually the direction becomes that of v_2 :

$$x(t) = e^{\lambda t} t \left(\frac{1}{t} c_1 v_1 + c_2 v_2 \right).$$

Then the trajectory depends on λ and differs from when he had two distinct eigenvectors for the $\lambda = 0$ case:

$$\begin{cases} |x(t)| \rightarrow \infty \text{ exponentially,} & \lambda > 0 \\ |x(t)| \rightarrow \infty \text{ linearly,} & \lambda = 0 \\ |x(t)| \rightarrow 0 \text{ exponentially,} & \lambda < 0 \end{cases}.$$

Aside 13.2.1. If we write $x_2 = x_2(x_1)$, how smooth is $x_2 \in C^k$, where $x_2 \notin C^{k+\epsilon}$?

13.3 Nonlinear Systems

Drawing phase portraits for nonlinear systems is hard. Consider f_1, f_2 polynomials and

$$\begin{aligned}\frac{d}{dt}x_1 &= f_1(x_1, x_2; b) \\ \frac{d}{dt}x_2 &= f_2(x_1, x_2; b).\end{aligned}$$

We have a critical point $(0,0)$ and other critical points X_c that are functions of the parameters.

Can we find a trajectory and parameter b that connects $(0,0)$ to X_c in a C^∞ way? That's a hard problem. It relates to having

$$\frac{dx_2}{dx_1} = \frac{f_1}{f_2},$$

trying to solve with a degenerate point, then running into trouble because the function is not Lipschitz.

13.4 Linear Stability

Definition 13.4.1 (Stable solution). We say that the ODE is stable at x^* if for every $\epsilon > 0$, there exists $\delta > 0$ such that for any x_0 ,

$$|x_0 - x^*| < \delta \implies |x(t) - x^*| < \epsilon$$

for every $t \geq t_0$. In other words, if we start close to the point at which the ODE is stable, it stay within an ϵ -ball forever. x^* is unstable if it is not stable.

Definition 13.4.2 (Asymptotically stable solution). We say that x^* is an asymptotically stable solution to the ODE if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$|x_0 - x^*| < \delta \implies \lim_{t \rightarrow \infty} x(t) = x^*$$

for every $t \geq t_0$.

Asymptotic stability implies stability. Stability allows for very small oscillations in the limit, while asymptotic stability requires convergence.

Consider our linear 2×2 system again:

$$\frac{d}{dt}X = AX.$$

Is it stable at $(0,0)$? We see that

- If $\operatorname{Re}(\lambda_i) < 0$ for $i = 1, 2$, then the ODE is stable and asymptotically stable (at $(0,0)$)

- If there exists $i \in \{1, 2\}$ such that $\operatorname{Re}(\lambda_i) > 0$, then since $|e^{\lambda_i t}| \rightarrow \infty$, the ODE is unstable (consider the system starting near x^* but not on the saddle path).
- If $\operatorname{Re}(\lambda_i) \leq 0$ for every i and there exists a particular i where $\operatorname{Re}(\lambda_i) = 0$, the system is not asymptotically stable. If $\lambda_1 = \lambda_2$, then the system is unstable. However, if $\lambda_1 \neq \lambda_2$, the system is stable. If we have the generalized eigenvector case, then the ODE is not stable at $(0, 0)$.

We generalize this to an arbitrary $n \times n$ matrix.

Theorem 13.4.3. (Linear Stability) Let $x^* = 0$ and consider

$$\frac{d}{dt}x = Ax.$$

Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of A .

- (i) If $\operatorname{Re}(\lambda_i) < 0$ for every i , then x^* is asymptotically stable and stable.
- (ii) If there exists an i such that $\operatorname{Re}(\lambda_i) > 0$, then x^* is unstable.
- (iii) If there exists i such that $\operatorname{Re}(\lambda_i) \geq 0$, then x^* is not asymptotically stable.
- (iv) If $\operatorname{Re}(\lambda_i) \leq 0$ for every i and the Jordan block J_i associated with λ_i for $\operatorname{Re}(\lambda_i) = 0$ has size 1×1 , then x^* is stable. If it has size $n \times n$ for $n \geq 2$, then x^* is unstable.

Furthermore, if there exists $\Lambda > 0$ such that

$$\operatorname{Re}(\lambda_i) \leq -\Lambda < 0$$

for every i , then we have for the solution $x(t) = e^{At}x_0$ that for any $\alpha \in (0, \Lambda)$

$$\max_i |x_i(t)| \leq Ce^{-\alpha t} \max_i |x_{0,i}|.$$

Notation 13.4.4. For a vector x and matrix M , define the norms

$$\|x\| = \max_{1 \leq i \leq n} |x_i|, \quad \|M\| = \max_{i,j} |M_{ij}|.$$

For a complex number $a + ib$, define the norm

$$|a + ib| = \sqrt{a^2 + b^2}.$$

WHY DO WE ONLY CARE ABOUT THE REAL PART OF THE EIGENVALUES IN OUR THEOREM STATEMENT? Consider the $n = 1$ case. We can write the solution to the ODE

$$x(t) = e^{(\alpha + i\beta)t}x_0,$$

which yields

$$|x(t)| = e^{\alpha t} |e^{i\beta t}| |x_0| = e^{\alpha t} |x_0|.$$

We see immediately that

$$\begin{cases} \alpha > 0 \implies |x(t)| \rightarrow \infty \implies \text{unstable} \\ \alpha = 0 \implies |x(t)| = |x_0| \implies \text{stable} \\ \alpha < 0 \implies |x(t)| \rightarrow 0 \implies \text{asymptotically stable} \end{cases}.$$

Now consider the $n \times n$ case, where we can write $A = UJU^{-1}$ for J being a matrix of Jordan blocks. Then

$$e^{At} = Ue^{Jt}U^{-1}, \quad e^{Jt} = \begin{pmatrix} e^{J_1 t} & & \\ & \ddots & \\ & & e^{J_m t} \end{pmatrix}.$$

Proof. (i) Suppose there is some $\lambda > 0$ such that $\operatorname{Re}(\lambda_i) \leq -\lambda < 0$ for every i . A may not be diagonalizable, so we consider its Jordan normal form. We know that for each block J_i ,

$$e^{J_i t} = e^{\lambda_i t} \begin{pmatrix} 1 & t & \dots & \frac{t^{n_i-1}}{(n_i-1)!} \\ & 1 & t & \dots \\ & & \ddots & \\ & & & 1 \end{pmatrix}.$$

Since exponential growth outpaces polynomial growth,

$$|e^{\lambda_i t} t^m| \leq |e^{\operatorname{Re}(\lambda_i) t} t^m| \leq C_A e^{-\frac{2}{3}\lambda t}$$

for $m \leq n_i \leq n$.

Since J is block diagonal, each Jordan block J_i satisfies

$$\|e^{J_i t}\| \leq C_i e^{-\frac{2}{3}\lambda t}.$$

Hence there is a constant $C_J > 0$ such that

$$\|e^{Jt}\| \leq C_J e^{-\frac{2}{3}\lambda t}.$$

Since $A = UJU^{-1}$, we have

$$e^{At} = Ue^{Jt}U^{-1},$$

and therefore

$$\|e^{At}\| \leq \|U\| \|e^{Jt}\| \|U^{-1}\| \leq \|U\| \|U^{-1}\| C_J e^{-\frac{2}{3}\lambda t} =: C_A e^{-\frac{2}{3}\lambda t}.$$

Finally, we see

$$|x(t)| = |e^{At} x_0| \leq C_A e^{-\frac{2}{3}\lambda t} |x_0|.$$

For any $m \in \mathbb{Z}$ and $\epsilon > 0$, there exists $C_{m,\epsilon}$ where $t^m \leq C_{m,\epsilon} e^{\epsilon t}$. We take the max of these from $m = 0, \dots, n_i - 1$ as "the" C_A .

Here, we took $\epsilon = \frac{1}{3}\lambda$ in $|t^m| \leq e^{\epsilon t} C_{m,\epsilon}$. Throughout this proof, the C_A s are changing because we are lazy.

This implies that for any x_0 , $x(t) \rightarrow 0$.

(ii) Suppose there exists i such that $\operatorname{Re}(\lambda_i) > 0$. Then

$$Av_i = \lambda_i v_i$$

tells us

$$x(t) = Ce^{\lambda_i t} v_i.$$

Here, we start the system at Cv_i , so it moves in the same direction.

We know that $\operatorname{Re}x(t)$ and $\operatorname{Im}x(t)$ solve the ODE. We saw earlier that

$$|x(t)| = |Cv_i| e^{\operatorname{Re}(\lambda_i)t} \rightarrow \infty$$

by assumption. Hence, the real and/or imaginary part of $x(t)$ must go to infinity. Since we can take a small perturbation in the real direction of v_i by starting at $\delta \operatorname{Re}(v_i)$, x^* must be unstable.

(iii) Suppose there exists i such that $\operatorname{Re}(\lambda_i) \geq 0$. Using the same ansatz,

$$|x(t)| = |Cv_i| e^{\operatorname{Re}(\lambda_i)t} \geq |Cv_i| > 0.$$

Hence x^* cannot be asymptotically stable.

(iv). Write $A = UJU^{-1}$ with J in Jordan form. Fix a Jordan block J_i associated with an eigenvalue λ such that $\Re(\lambda) = 0$.

As an application, consider

$$\frac{d}{dt}x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} x$$

or

$$\frac{d}{dt}x = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} x.$$

Then $\lambda = \pm i$, so the system just rotates.

STEP 1 (EXPLICIT FORM). Let the size of J_i be $m \times m$ and write

$$J_i = \lambda I + N,$$

where N is nilpotent with $N^m = 0$ and $N^{m-1} \neq 0$. Then

$$e^{J_i t} = e^{\lambda t} \sum_{k=0}^{m-1} \frac{t^k}{k!} N^k.$$

STEP 2 (POLYNOMIAL GROWTH WHEN $m \geq 2$). Let e_m be the last standard basis vector in the coordinates of this block. Since $N^{m-1}e_m = e_1$,

$$\left(e^{J_i t} e_m \right)_1 = e^{\lambda t} \frac{t^{m-1}}{(m-1)!}.$$

Hence, in the max norm,

$$\|e^{J_i t} e_m\| \geq \frac{t^{m-1}}{(m-1)!}.$$

Let y_0 be the vector in $\mathbb{R}^n$ that equals e_m in the i th block and 0 elsewhere. Then $\|y_0\| = 1$ and

$$\|e^{J_i t} y_0\| \geq \frac{t^{m-1}}{(m-1)!}.$$

Set $x_0 = \eta U y_0$. Then

$$\|x_0\| \leq \eta \|U\|, \quad \|x(t)\| = \|U e^{J_i t} (\eta y_0)\| \geq \frac{\eta}{\|U^{-1}\|} \|e^{J_i t} y_0\| \geq \frac{\eta}{(m-1)! \|U^{-1}\|} t^{m-1}.$$

Given any $\delta > 0$, choose $\eta < \delta / \|U\|$. Then $\|x_0\| < \delta$, but choosing t large makes $\|x(t)\|$ arbitrarily large. Thus $x^* = 0$ is unstable when $m \geq 2$.

STEP 3 (BOUNDEDNESS WHEN $m = 1$). If every Jordan block with $\Re(\lambda) = 0$ has size 1×1 , then $e^{J_i t} = e^{\lambda t}$ on those blocks, so $\|e^{J_i t}\| = 1$, while blocks with $\Re(\lambda) < 0$ satisfy $\|e^{J_i t}\| \rightarrow 0$. Hence $\sup_{t \geq 0} \|e^{J t}\| < \infty$, so $\sup_{t \geq 0} \|e^{A t}\| < \infty$, which implies stability of 0.

□

13.5 Nonlinear Stability

Consider $F \in C^2$ such that $F(x^*) = 0$ and we have

$$\frac{d}{dt}x = F(x).$$

We can Taylor expand:

$$\frac{d}{dt}x = F(x) = F(x^*) + \nabla F(x^*)(x - x^*) + G(x - x^*).$$

If $|x - x^*| < 1$, then

$$G(x - x^*) \leq M |x - x^*|^2.$$

To simplify notation, write

$$z(t) = x(t) - x^*$$

and

$$A = \nabla F(x^*).$$

Since, $F(x^*) = 0$, we get

$$\frac{d}{dt}z(t) = \frac{d}{dt}x(t) = Az + G(z), \quad |G(z)| \leq M |z|^2, \quad |z| \leq 1.$$

Because x^* was an equilibrium for the ODE in x , we know that 0 is an equilibrium of the ODE in z .

Theorem 13.5.1. (Nonlinear stability) Suppose we have

$$\frac{d}{dt}\vec{x} = F(x)$$

where $F \in C^1$ and let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of $A = \nabla F(x^*)$.⁵

(i) If $\Re(\lambda_i) < 0$ for every i , then x^* is asymptotically stable and stable.

(ii) If there exists i such that $\Re(\lambda_i) > 0$, then x^* is unstable.

By the bound on G , we conjecture that close to 0, $\frac{d}{dt}z$ behaves like Az .

He wrote $F \in C^2$ on the board, but I think only C^1 is necessary (Braun and his handwritten notes confirm this).

⁵ Really, this should be $A = J_F(x^*)$.

Otherwise (when $\operatorname{Re}(\lambda_i) = 0$), the behavior is not clear and we need to know more about G .⁶

Proof. (i) Assume that for every i , $\operatorname{Re}(\lambda_i) \leq -\lambda < 0$. To generalize for equilibria not at the origin, we write $z = x - x^*$. As hinted above, we conjecture that for z close to 0,

$$\frac{d}{dt}z \approx Az,$$

which would imply

$$|z(t)| \approx \left| e^{At} z_0 \right| \leq C_A e^{-\frac{2}{3}\lambda t} \underbrace{\left| z_0 \right|}_{\delta}.$$

This is because by our assumption that the real parts of all eigenvalues of A are less than $-\lambda$, for any $\alpha \in (0, \lambda)$, we have that

$$\|e^{At}\| \leq C_A e^{-\alpha t}$$

for some constant C_A . We just pick $\alpha = \frac{2}{3}\lambda$ for convenience.

Correspondingly, with this approximation,

$$\begin{aligned} |G(z(t))| &\leq M |z(t)|^2 \\ &\leq M C_A^2 \delta^2 e^{-\frac{4}{3}\lambda t}. \end{aligned}$$

We want to show that if $|z_0| = \delta \ll 1$, then

$$d(t) := e^{\frac{2}{3}\lambda t} |z(t)| \leq \tilde{C}_A \delta.$$

STEP 1 (LOCAL EXISTENCE). Since F is locally Lipschitz, the solution with $z(0) = z_0$ exists on some $[0, T)$.

STEP 2 (DECOMPOSITION). Write $z(t) = x(t) - x^*$ then define

$$A := \nabla F(x^*), \quad G(z) := F(x^* + z) - Az.$$

Then $G(0) = 0$ and

$$z'(t) = x'(t) = Az(t) + G(z(t)).$$

Assume (e.g. if $F \in C^2$, which makes things simpler) that there exist $M, \delta > 0$ such that

$$\|G(z)\| \leq M \|z\|^2 \quad \text{whenever } \|z\| < \delta.$$

STEP 3 (DUHAMEL BOUND). Since $\Re(\lambda_i) \leq -\lambda$, there exists $C_A \geq 1$ such that

$$\|e^{At} v\| \leq C_A e^{-\frac{2}{3}\lambda t} \|v\| \quad \forall t \geq 0.$$

⁶ To see why $\operatorname{Re}(\lambda_i) = 0$ complicates things, consider

$$\frac{d}{dt}x = 0 + x^2.$$

$x^* = 0$ is an equilibrium. If we perturb with $x(0) = \epsilon$ for $\epsilon > 0$, then we get finite-time blowup. But if $x(0) = -\epsilon$, we have stability.

We can move the factor on $|z(t)|$ to the right hand side to get the approximate inequality we expect, which will show asymptotic stability.

By variation of constants,

$$z(t) = e^{At}z_0 + \int_0^t e^{A(t-s)}G(z(s))ds.$$

Hence

$$\begin{aligned}\|z(t)\| &\leq C_A e^{-\frac{2}{3}\lambda t} \|z_0\| + C_A \int_0^t e^{-\frac{2}{3}\lambda(t-s)} \|G(z(s))\| ds \\ &\leq C_A e^{-\frac{2}{3}\lambda t} \delta + \tilde{C}_A \int_0^t e^{-\frac{2}{3}\lambda(t-s)} \|z(s)\|^2 ds,\end{aligned}$$

where $\delta := \|z_0\|$.

Let

$$d(t) := e^{\frac{2}{3}\lambda t} \|z(t)\|.$$

Then

$$d(t) \leq C_A \delta + \tilde{C}_A \int_0^t e^{-\frac{2}{3}\lambda s} d(s)^2 ds.$$

STEP 4 (BOOTSTRAP). Assume $d(s) \leq 2C_A \delta$ on $[0, t]$, which is feasible since $d(t) = 0$. Plugging this into the right hand side of the previous inequality,

$$d(t) \leq C_A \delta + \tilde{C}_A (2C_A)^2 \delta^2 \int_0^t e^{-\frac{2}{3}\lambda s} ds \leq C_A \delta + C_{A,2} \delta^2.$$

For δ small enough, $d(t) \leq 2C_A \delta$. Substituting this into the definition of $d(t)$ yields

$$\|z(t)\| \leq 2C_A \delta e^{-\frac{2}{3}\lambda t} \rightarrow 0,$$

so x^* is asymptotically stable.

(ii) Homework. □

13.6 Lyapunov Stability

Consider

$$\frac{d}{dt}x = F(x), \quad x, F(x) \in \mathbb{R}^{n \times 1}.$$

We showed last time that

- (i) If $\frac{d}{dt}x = Ax$ has equilibrium x^* , we can put $z(t) = x(t) - x^*$ to get

$$\frac{d}{dt}z(t) = Az(t) + G(z).$$

In this case, stability relates to the eigenvalues of $A = \nabla F(x^*)$.

Example 13.6.1. (Pendulum system) We have

$$m \frac{d^2\theta}{dt^2} + \frac{mg}{L} \sin(\theta) = 0.$$

Putting $x = \theta, y = \theta'$,

$$\frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ -\frac{g}{L} \sin(x) \end{pmatrix}.$$

We see the equilibria are in the form $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} k\pi \\ 0 \end{pmatrix}$.

Denote “even” θ as $\theta_e = (2k\pi, 0)$ and “odd” as $\theta_o = (2k\pi + \pi, 0)$. We have that

$$F = \begin{pmatrix} y \\ -\frac{g}{L} \sin(x) \end{pmatrix},$$

so

$$\begin{aligned} J_F|_{x^*} &= \begin{pmatrix} \partial_x F_1 & \partial_y F_1 \\ \partial_x F_2 & \partial_y F_2 \end{pmatrix} \Big|_{x^*} \\ &= \begin{pmatrix} 0 & 1 \\ -\frac{g}{L} \cos(x) & 0 \end{pmatrix}. \end{aligned}$$

For

$$A_o = J_F|_{\theta_o} = \begin{pmatrix} 0 & 1 \\ \frac{g}{L} & 0 \end{pmatrix},$$

we get that

$$\lambda^2 = \frac{g}{L} \implies \lambda_{\pm} = \pm \sqrt{\frac{g}{L}},$$

so $\lambda_- < 0 < \lambda_+$. Then the odd solutions are unstable.

For

$$A_e = J_F|_{\theta_e} = \begin{pmatrix} 0 & 1 \\ -\frac{g}{L} & 0 \end{pmatrix},$$

we get that

$$\lambda^2 = \pm i \sqrt{\frac{g}{L}} \implies \operatorname{Re}(\lambda) = 0.$$

Thus the system might be stable.

In the linearized method for nonlinear systems, we only had results for small perturbations $|x_0 - x^*| \ll 1$, since we used a Taylor expansion. With the Lyapunov method⁷, we can have a large range of initial perturbation. We don’t get this advantage for free, though; while the linearized method can apply to any autonomous and C^1 F , while the Lyapunov method only applies to a small set of systems, namely, those where we can actually find a Lyapunov function.

Definition 13.6.2 ($\dot{L}$ along trajectories). Let $O \subseteq \mathbb{R}^n$ be open and let $F : O \rightarrow \mathbb{R}^n$ be locally Lipschitz. Consider the autonomous ODE

$$\frac{d}{dt}x(t) = F(x(t)), \quad x(t) \in O.$$

We can’t say that the system is definitely stable given what we know because the system is not linear.

⁷ Also called global method

Let $L : O \rightarrow \mathbb{R}$ be of class C^1 .

For any solution $x(\cdot)$ with values in O , the derivative of L along the trajectory is given by the chain rule:

$$\frac{d}{dt}L(x(t)) = \nabla L(x(t)) \cdot \dot{x}(t) = \nabla L(x(t)) \cdot F(x(t)).$$

The function

$$\dot{L}(x) = F(x) \cdot \nabla L(x), \quad x \in O,$$

is called the derivative of L along the vector field F .

$\frac{d}{dt}L(x(t))$ is the derivative of the distance function along the particular solution $x(t)$, while $\dot{L}(x)$ is defined on the state space O before a solution $x(\cdot)$ is picked. On the solution, $\frac{d}{dt}L(x(t))$ and $\dot{L}(x)$ coincide.

Theorem 13.6.3 (Lyapunov Stability). *Let $O \subseteq \mathbb{R}^n$ be open and let $F : O \rightarrow \mathbb{R}^n$ be locally Lipschitz. Consider the autonomous system*

$$\dot{x}(t) = F(x(t)).$$

Let $x^ \in O$ be an equilibrium, i.e. $F(x^*) = 0$.*

Suppose there exists x^ a function L such that*

- (i) *Positive definiteness: $L(x^*) = 0$ and $L(x) > 0$ for every $x \in O \setminus \{x^*\}$.*
- (ii) *Sign of the orbital derivative: $\dot{L}(x) \leq 0$ for every $x \in O \setminus \{x^*\}$.*

Then x^ is (Lyapunov) stable.*

Furthermore, if in (ii) we have the strict inequality

$$\dot{L}(x) < 0 \quad \text{for every } x \in O \setminus \{x^*\},$$

then x^ is asymptotically stable.*

Proof. Let $x(t; x_0)$ denote the solution with initial condition $x(0; x_0) = x_0$. Since F is locally Lipschitz, solutions exist and are unique at least locally in time.

(STEP 1) Derivative of L along trajectories. For any solution $x(t) = x(t; x_0)$ with values in O , the chain rule gives

$$\frac{d}{dt}L(x(t)) = \nabla L(x(t)) \cdot \dot{x}(t) = \nabla L(x(t)) \cdot F(x(t)) = \dot{L}(x(t))$$

with the dot product. Thus $\frac{d}{dt}L(x(t)) \leq 0$ (or < 0) whenever $x(t) \neq x^*$.

(STEP 2) Stability. Let $\delta > 0$ be arbitrary with $\overline{B(x^*, \delta)} \subset O$. Consider the sphere

$$S_\delta := \{y \in \mathbb{R}^n : |y - x^*| = \delta\}.$$

Think of L as a generalized distance to some point x^* , e.g. $L(x) = \|x - x^*\|$.

L is called a Lyapunov candidate when we first propose it and is called a Lyapunov function once we show that it satisfies the positive definiteness and sign condition on its derivative.

In standard notation, we take $x \in \mathbb{R}^n$ to be a column vector, so $\dot{x}$ and $\nabla L(x(t))$ are also column vectors.

By positive definiteness of L , $L > 0$ on S_δ , and S_δ is compact. Hence by continuity of L ,

$$\alpha := \min_{y \in S_\delta} L(y) > 0.$$

By continuity of L at x^* and $L(x^*) = 0$, there exists $\varepsilon \in (0, \delta)$ such that

$$|x - x^*| < \varepsilon \Rightarrow L(x) < \alpha.$$

Now take any initial condition x_0 with $|x_0 - x^*| < \varepsilon$, and let $x(t) = x(t; x_0)$ be the corresponding solution (defined on its maximal interval $[0, T_{\max})$). As long as $x(t) \in B(x^*, \delta)$, we have $\frac{d}{dt}L(x(t)) \leq 0$, hence

$$L(x(t)) \leq L(x_0) < \alpha.$$

We claim that $x(t)$ never reaches the sphere S_δ . Indeed, if there were $t_1 > 0$ with $|x(t_1) - x^*| = \delta$, then $x(t_1) \in S_\delta$, so $L(x(t_1)) \geq \alpha$, contradicting $L(x(t_1)) \leq L(x_0) < \alpha$. Therefore

$$|x(t) - x^*| < \delta \quad \text{for all } t \in [0, T_{\max}).$$

This shows: for the given $\delta > 0$ there exists $\varepsilon > 0$ such that $|x_0 - x^*| < \varepsilon$ implies $|x(t; x_0) - x^*| < \delta$ for all times for which the solution exists. Since the solution never leaves the compact set $\overline{B(x^*, \delta)}$, standard continuation results for ODEs imply $T_{\max} = +\infty$. Hence the definition of Lyapunov stability is satisfied.

(STEP 3) Now assume the strengthened condition $\dot{L}(x) < 0$ for all $x \neq x^*$. We already know from Step 2 that $x(t)$ remains in a compact ball $K = \overline{B(x^*, \delta)} \subset O$ for all $t \geq 0$, so the solution exists for all $t \geq 0$ and $L(x(t))$ is monotone nonincreasing and bounded below by 0. Thus the limit

$$\ell := \lim_{t \rightarrow \infty} L(x(t))$$

exists.

Suppose for contradiction that $x(t)$ does not converge to x^* . Then there exists a point $z^* \in K$ with $z^* \neq x^*$ and a sequence $t_n \rightarrow \infty$ such that

$$x(t_n) \rightarrow z^*.$$

By continuity of L ,

$$L(x(t_n)) \rightarrow L(z^*),$$

and since the limit exists, we have

$$\ell = \lim_{t \rightarrow \infty} L(x(t)) = \lim_{n \rightarrow \infty} L(x(t_n)) = L(z^*) > 0.$$

For any $t \geq 0$, since $\dot{L}(x(t)) \leq 0$ along the trajectory, L is nonincreasing in t . Hence whenever $t_m > t$ we have

$$L(x(t_m)) \leq L(x(t)).$$

Because $x(t_n) \rightarrow z^*$ and L is continuous, we also have $L(x(t_n)) \rightarrow L(z^*)$. Thus, for fixed $t \geq 0$ and any $\epsilon > 0$ we can choose m large enough such that $t_m > t$ and

$$L(x(t_m)) > L(z^*) - \epsilon.$$

Combining, we obtain

$$L(x(t)) \geq L(x(t_m)) > L(z^*) - \epsilon.$$

Since $\epsilon > 0$ is arbitrary, it follows that

$$L(x(t)) \geq L(z^*) > 0 \quad \text{for all } t \geq 0. \quad (32)$$

Thus $L(z^*)$ is a lower bound for all values of L along the trajectory starting at x_0 .

Now define

$$\dot{z}(t) = F(z(t)), \quad z(0) = z^*.$$

By assumption (ii), $\dot{L}(x) < 0$ for $x \neq x^*$, so in particular $\dot{L}(z(t)) < 0$ for $t > 0$. Hence $L(z(t))$ is strictly decreasing and

$$\Delta := L(z^*) - L(z(1)) = L(z(0)) - L(z(1)) > 0.$$

Next, for each n , let $y_n(t)$ solve

$$\dot{y}_n(t) = F(y_n(t)), \quad y_n(0) = x(t_n),$$

so that by uniqueness $y_n(1) = x(t_n + 1)$. Since $x(t_n) \rightarrow z^*$ and solutions depend continuously on initial data, we have $y_n(1) \rightarrow z(1)$ as $n \rightarrow \infty$. On the compact ball that contains the trajectory, L is Lipschitz, so there exists $C > 0$ such that

$$|L(y_n(1)) - L(z(1))| \leq C |y_n(1) - z(1)| \rightarrow 0,$$

so $L(y_n(1)) \rightarrow L(z(1))$.

We now compute

$$\begin{aligned} L(x(t_n + 1)) - L(z^*) &= L(y_n(1)) - L(z^*) \\ &= (L(y_n(1)) - L(z(1))) + (L(z(1)) - L(z^*)) \\ &= (L(y_n(1)) - L(z(1))) - \Delta. \end{aligned}$$

Since $L(y_n(1)) \rightarrow L(z(1))$, we can choose n large enough that $|L(y_n(1)) - L(z(1))| < \Delta/2$, and hence

$$L(x(t_n + 1)) - L(z^*) \leq \frac{\Delta}{2} - \Delta = -\frac{\Delta}{2} < 0,$$

so $L(x(t_n + 1)) < L(z^*)$.

However, applying (32) with $t = t_n + 1$ gives $L(x(t_n + 1)) \geq L(z^*)$, a contradiction. Thus our assumption that $x(t) \not\rightarrow x^*$ was false, and therefore $x(t) \rightarrow x^*$ as $t \rightarrow \infty$. $\square$

We know that for all $t > 0$, $z(t) \neq x^*$: if $z(t_0) = x^*$ for some $t_0 > 0$, then both the constant solution $x(t) \equiv x^*$ and $z(t)$ solve $\dot{x} = F(x)$ with $x(t_0) = x^*$, contradicting uniqueness unless $z \equiv x^*$, which is impossible since $z(0) = z^* \neq x^*$.

Continuous dependence tells us that if the initial points $x(t_n)$ converge to z^* , then for each fixed t the solutions $x(t; x(t_n))$ converge to $x(t; z^*)$; in particular $y_n(1) = x(1; x(t_n)) \rightarrow x(1; z^*) = z(1)$. This is because for the same time t , we have

$$|y_n(t) - z(t)| \leq e^{Lt} |y_0 - z_0| = e^{Lt} |x(t_n) - z^*|.$$

Example 13.6.4. Considering the pendulum from before with L as the length of the pendulum, the total energy of the system is

$$L(x, y) = \frac{1}{2}mL^2\dot{y}^2 + mgL(1 - \cos(x)).$$

We claim that L is a Lyapunov function for θ_e . Note that $L(0, 0) = 0$.

Furthermore, $L(x, y) > 0$ when $(x, y) \neq (0, 0)$. Specify $O = (-\pi, \pi) \times \mathbb{R}$, so the assumption on $x \neq x^*$ is satisfied.

Computing

$$\begin{aligned}\dot{L} &= \nabla L \cdot F \\ \nabla L &= \begin{pmatrix} mgL \sin(x) & mL^2\dot{y} \end{pmatrix}, \quad F = \begin{pmatrix} y \\ -\frac{g}{L} \sin(x) \end{pmatrix} \\ \dot{L} &= mgL \sin(x)y - mL^2\dot{y} \frac{g}{L} \sin(x) = 0,\end{aligned}$$

we see that the total energy is always constant (since $\dot{L} = 0$). Thus, the system is stable only if $L(x, y) = 0$, i.e. if $(x, y) = (0, 0)$. If $L(x(t)) = L(x_0) \neq 0$, then L does not go to 0 and just stays at $L(x_0)$ forever.

Example 13.6.5. Let

$$\begin{aligned}x' &= -x^3 \\ y' &= -y(x^2 + z^2 + 1) \\ z' &= -\sin(z).\end{aligned}$$

We have one equilibrium in $(x, y, z) = (0, 0, 0)$.

Define $v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $L(v) = x^2 + y^2 + z^2$, which is continuous and in particular, Lipschitz.

Then

$$\nabla L \cdot F = 2 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot F = -2x^4 - 2y^2(x^2 + z^2 + 1) - 2z \sin(z).$$

We want to find some domain $O \setminus \{0\}$ where this quantity is nonpositive (to show stability). Since $-2x^4 - 2y^2(x^2 + z^2 + 1) < 0$ for all v , we just need to worry about $2z \sin(z)$, which is strictly negative when $z \in (-\pi, \pi)$.

Then

$$\nabla L \cdot F < 0$$

for all $v \neq (0, 0, 0)$, which is assumption (ii). Hence the system is asymptotically stable for any $x \in O \setminus x^*$.

Remark 13.6.6. When we have to construct a Lyapunov candidate, try ones with physical connections to energy equations or the square of Euclidean distance.⁸ Another useful candidate if $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is

Refer to BDM 9.6 for an example and Smale 9.2 for a proof.

⁸ If the candidate uses distance, it should be distance to the equilibrium — not necessarily the origin!

$$L(x, y) = ax^2 + bxy + cy^2$$

for constants a, b, c . With this candidate, it is important to know that L is positive definite iff

$$a > 0, \quad 4ac - b^2 > 0,$$

and negative definite iff

$$a < 0, \quad 4ac - b^2 > 0.$$

13.7 Signing the Lyapunov Equation

Example 13.7.1. Suppose

$$\frac{d}{dt} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} (\epsilon x)(z+1) \\ (-x + \epsilon y)(z+1) \\ -z^2 \end{pmatrix} = F(v).$$

The only equilibrium is $x^* = (0, 0, 0)$.

Using the method of Taylor expansion,

$$\begin{aligned} \frac{d}{dt}v &= Av + o(|v|^2) \\ A = \nabla F|_{x^*} &= \begin{pmatrix} \epsilon & 2-1 & \epsilon & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

$$\lambda \left((\lambda - \epsilon)^2 + 2 \right) = 0$$

$$\lambda = 0, \epsilon \pm \sqrt{2}i,$$

and the system is unstable for $\epsilon > 0$. What about when $\epsilon \leq 0$?

Try

$$L = ax^2 + by^2 + cz^2$$

for $a, b, c > 0$. We see $L(v^*) = 0$ and $L(v) > 0$ for all $v \neq v^*$.

For the first assumption, note

$$\begin{aligned} \nabla L \cdot F &= \begin{pmatrix} 2ax \\ 2by \\ 2cz \end{pmatrix} \cdot \begin{pmatrix} (\epsilon x + 2y)(z+1) \\ (-x + \epsilon y)(z+1) \\ -z^3 \end{pmatrix} \\ &= (z+1) [2ax(\epsilon x + 2y) + 2by(-x + \epsilon y) - 2cz^4] \\ &= (z+1) \left[\underbrace{2a\epsilon x^2}_{\leq 0} + \underbrace{2b\epsilon y^2}_{\leq 0} + (4a - 2b)xy \right]. \end{aligned}$$

Use one of our degrees of freedom to require $4a - 2b = 0$ with $a = 1, b = 2, c = 1$. Since $z + 1 > 0$ for $z \in [-1, \infty)$, we put $O = \mathbb{R} \times \mathbb{R} \times (-1, \infty)$ to get that

$$\epsilon = 0: \quad \dot{L} \leq 0 \text{ on } O \implies \text{stable}$$

$$\epsilon < 0: \quad \dot{L} < 0 \text{ on } O \setminus \{x_0\} \implies \text{asymptotically stable.}$$

If $(x^*, y^*, z^*) \neq 0$, we could define this as

$$L = a(x - x^*)^2 + b(y - y^*)^2 + c(z - z^*)^2.$$

Note that $B(0,1) \subseteq O$. Then with our function $L = x^2 + 2y^2 + z^2$,

$$\alpha = \min_{\partial B(0,1)} L(x) = 1.$$

If $x \in B(0, \frac{1}{\sqrt{2}})$, then $L(x) \leq 2(x^2 + y^2 + z^2) < 2 \cdot \frac{1}{2} = 1$.

Hence, $x_0 \in B(0, \frac{1}{\sqrt{2}})$ implies that $x(t) \in B(0,1)$ for every $t > 0$.

what are we doing

Example 13.7.2. Consider

$$ax^2 + bxy + cy^2.$$

This is positive for any $(x,y) \neq (0,0)$ when $a, c > 0$ and $4ac > b^2$.

This is obtained by putting $t = \frac{x}{y}$ and finding the minimum of $at^2 + bt + c$ and requiring it to be positive.

Example 13.7.3. Consider

$$\begin{aligned} x' &= -x - xy^2 \\ y' &= -y - x^2y + Mx. \end{aligned}$$

The equilibrium if $(0,0)$. Linearizing, we get

$$\frac{d}{dt}x = \begin{pmatrix} -1 & 0 \\ M & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + o(|x|^2).$$

$\lambda = -1$, so the system is asymptotically stable.

With the Lyapunov method, we put

$$L = x^2 + cy^2$$

so

$$\dot{L} = \nabla L \cdot F = \begin{pmatrix} 2x \\ 2cy \end{pmatrix} \begin{pmatrix} -x - xy^2 \\ -y - x^2y + Mx \end{pmatrix}.$$

The conditions for nonpositivity are

$$x^2 + cy^2 - cMxy > 0$$

for any $(x,y) \neq (0,0)$, or that

$$\frac{4}{M^2} > c > 0$$

gives a valid Lyapunov function in $\mathbb{R} \setminus \{x^*\}$.

Remark 13.7.4. Signing $\dot{L} = \nabla L \cdot F$ is difficult for most F . This is why the linearization method is much more general.

14 Dynamical Systems

14.1 Introduction to Dynamical Systems

Consider the autonomous system

$$\frac{d}{dt}x = F(x), \quad x \in \mathbb{R}^n.$$

Let $\phi_t(x)$ denote the solution at time t with initial condition $\phi_0(x) = x$. It is the value of the ODE t units of time after it passes x .

If for some $t_0 \in \mathbb{R}$ and $T > 0$ we have

$$x(t_0 + T) = x(t_0),$$

then by existence and uniqueness,

$$x(s + T) = x(s) \quad \text{for all } s \in \mathbb{R},$$

so the trajectory through $x(t_0)$ is periodic with (minimal) period T .

Notation 14.1.1. For the system

$$\frac{d}{dt}x = F(x),$$

we define the forward and backward orbits (trajectories) of a point x by

$$\gamma_+(x) = \{\phi_t(x) : t \geq 0, \phi_0(x) = x\},$$

$$\gamma_-(x) = \{\phi_t(x) : t \leq 0, \phi_0(x) = x\},$$

where $\phi_t(x)$ is the value at time t of the solution for the ODE with the initial condition $\phi_0(x) = x$.

The ω -limit set of x is

$$\omega(x) = \{Y : \exists t_n \rightarrow +\infty \text{ with } \phi_{t_n}(x) \rightarrow Y\},$$

and the α -limit set of x is

$$\alpha(x) = \{Y : \exists t_n \rightarrow -\infty \text{ with } \phi_{t_n}(x) \rightarrow Y\}.$$

A set D is positively invariant if for every $t \geq 0$,

$$x \in D \implies \phi_t(x) \in D,$$

and negatively invariant if for every $t \leq 0$,

$$x \in D \implies \phi_t(x) \in D.$$

By autonomy of the system, the flow satisfies the semigroup property

$$\phi_{t+s}(x) = \phi_t(\phi_s(x)) \quad \text{for all } s, t \in \mathbb{R},$$

and in particular

$$x = \phi_{t+(-t)}(x) = \phi_t(\phi_{-t}(x)).$$

We say that the ω -limit set of x_0 is a closed orbit if there exists a periodic solution $x(t)$ with period $T > 0$ such that

$$\omega(x_0) = \{x(t) : 0 \leq t < T\}.$$

Example 14.1.2. (Positive invariant set) Consider

$$\frac{d}{dt}X = \begin{pmatrix} -1 & \\ & -2 \end{pmatrix} X,$$

so

$$X(t) = \begin{pmatrix} e^{-t}X_{0,1} \\ e^{-2t}X_{0,2} \end{pmatrix}.$$

Then $B_R(0)$ is positive invariant.

Example 14.1.3. (Positive invariant set, homework) If $F(X) \cdot X|_{\partial B_R} > 0$, then B_R^c is positive invariant.

The flow points outwards on the boundary of the ball.

Theorem 14.1.4 (Poincaré–Bendixson). Let $\dot{x} = F(x)$ be an autonomous system on $\mathbb{R}^2$, and let $x(t) = \phi_t(x_0)$ be a solution whose forward orbit is contained in a bounded region. Suppose the ω -limit set $\omega(x_0)$ is nonempty, closed, and bounded, and that $\omega(x_0)$ contains no equilibrium points of the system. Then $\omega(x_0)$ is a periodic orbit; that is, it is a closed orbit of the ODE solution.

See Smale, Section 10.

Proposition 14.1.5. If x, z lie on the same trajectory, then $\omega(x) = \omega(z)$ and $\alpha(x) = \alpha(z)$.

Proof. Suppose $Y \in \omega(x)$. Since x and z are on the same trajectory, we can write

$$x = \phi_{s_1}(x_0), \quad z = \phi_{s_2}(x_0).$$

We can find $\{t_n\}$ such that $\phi_{t_n}(x) \rightarrow Y$, and we can also write

$$z = \phi_{s_2-s_1}(x), \quad x = \phi_{s_1-s_2}(z).$$

Making the substitutions,

$$\phi_{t_n}(\phi_{s_1-s_2}(z)) = \phi_{t_n+s_1-s_2}(z) = \phi_{t_n+s_1-s_2}(\phi_{s_2-s_1}(x)) = \phi_{t_n}(x) \rightarrow Y,$$

so $Y \in \omega(z)$. Since Y was arbitrary, $\omega(x) \subseteq \omega(z)$. By symmetry, $\omega(x) = \omega(z)$.

Similarly, we can show $\alpha(x) = \alpha(z)$. □

Proposition 14.1.6. If D is positive invariant and closed, and $z \in D$, then $\omega(z) \subseteq D$. In particular, if we define

$$\omega(D) := \bigcup_{z \in D} \omega(z),$$

then $\omega(D) \subseteq D$.

Proof. We can find $\{t_n\}$ such that

$$\phi_{t_n}(z) \rightarrow Y \in \omega(z).$$

D being closed implies that $Y \in D$. Hence, $\omega(z) \subseteq D$.

Fix $z \in D$ and let $Y \in \omega(z)$. By definition of the ω -limit set, there exists $\{t_n\}$ with $t_n \geq 0$ such that $\phi_{t_n}(z) \rightarrow Y$.

Since $z \in D$ and D is positively invariant, we have

$$\phi_t(z) \in D \quad \text{for all } t \geq 0.$$

In particular,

$$\phi_{t_n}(z) \in D \quad \text{for all } n.$$

Thus $\{\phi_{t_n}(z)\}$ is a sequence in D converging to Y . Because D is closed, any limit of a convergent sequence in D lies in D , so $Y \in D$.

Since $Y \in \omega(z)$ was arbitrary, we conclude $\omega(z) \subseteq D$. z was arbitrary, so $\bigcup_{z \in D} \omega(z) = \omega(D) \subseteq D$. $\square$

14.2 Proof of the Poincaré–Bendixson Theorem

Consider $\frac{d}{dt}X = F(X)$ where $X, F \in \mathbb{R}^{2 \times 1}$ and F is Lipschitz in any bounded domain D , where $|\nabla F| \leq L$ on D .

Recall that we have the semigroup property for $\phi_t(x)$, i.e.

$$\phi_t(\phi_s(x)) = \phi_{t+s}(x).$$

We also have continuous dependence: for $\phi_s(X), \phi_s(Y) \in D$, we get from Cornwall's inequality that

$$|\phi_t(X) - \phi_t(Y)| \leq e^{Lt} |X - Y|.$$

The Implicit Function Theorem says that for $F \in C^1$, $F(x_0, y_0) = 0$ such that $\partial_y F(x_0, y_0) \neq 0$, we can implicitly define a C^1 function $y := y(x)$ such that $F(x, y(x)) = 0$ for any $x \in B_\epsilon(x_0)$.

Theorem 14.2.1. (Poincaré–Bendixson) Suppose $\frac{d}{dt}X = F(X)$ is an autonomous ODE in $\mathbb{R}^2$ and that for some X_0 , $\omega(X_0)$ is bounded. Then $\omega(X_0)$ is either an equilibrium or contains a closed orbit (periodic solution).

We are examining some point X_0 where $F(X_0) \neq 0$. We want to reduce the 2D dynamics to 1D, as a bounded solution in 1D implies the existence of an equilibrium or a periodic solution since F is autonomous. After moving to 1D, we will prove that $S \cap \gamma_t(x)$ is monotone for some line S .

If $F(X_0) = 0$, then the behavior is trivial.

Remark 14.2.2. Fix X_0 with $F(X_0) \neq 0$. Choose a unit vector v_0 such that $v_0 \perp F(X_0)$. The idea is to take a short line segment through X_0 that is transverse to the flow: at X_0 the vector field points across this line rather than along it.

Define the line through X_0 in the direction v_0 by

$$L(X_0) = \{X_0 + tv_0 : t \in \mathbb{R}\}.$$

Equivalently, we parametrize this line by

$$h : \mathbb{R} \rightarrow L(X_0), \quad h(u) = X_0 + uv_0,$$

so $h(0) = X_0$.

Because F is continuous and $F(X_0) \neq 0$, the direction of the vector field cannot flip abruptly near X_0 . Hence there exists $\epsilon > 0$ such that

$$F(X) \cdot F(X_0) > 0 \quad \text{whenever } |X - X_0| < \epsilon.$$

Intuitively, this means that near X_0 the flow still points to the same side determined by $F(X_0)$.

Define the local section

$$S = \{h(u) \in L(X_0) : |u| \leq \epsilon\}.$$

Thus S is a small segment of the line orthogonal to $F(X_0)$ through X_0 .

Now define the flow-box map

$$\Psi : N_\sigma \rightarrow \Psi(N_\sigma) =: V_\sigma \quad \text{by} \quad \Psi(s, u) = \phi_s(h(u)),$$

where

$$N_\sigma = \{(s, u) : |s| < \sigma, |u| < \sigma\}.$$

The point of Ψ is that we first choose a starting point on the section S (using u), and then flow it forward or backward by time s . So Ψ builds a small “rectangle” of nearby trajectories around X_0 .

Note that $X_0 \in V_\sigma$. Moreover, for σ sufficiently small, Ψ is one-to-one since $J_\Psi(0,0) \neq 0$. Intuitively, this says that near $(0,0)$ the coordinates (s, u) provide a valid local description of points in the plane: each point in the flow box lies on exactly one nearby trajectory and meets the section at exactly one nearby point.

To make this explicit, take any $X \in V_\sigma$. Since $V_\sigma = \Psi(N_\sigma)$, we can write

$$X = \Psi(s_1, u_1) = \phi_{s_1}(h(u_1))$$

for some $(s_1, u_1) \in N_\sigma$. Then flowing backward by s_1 returns X to the section line:

$$\phi_{-s_1}(X) = \phi_{-s_1}(\phi_{s_1}(h(u_1))) = h(u_1) \in L(X_0).$$

In words: every point in the flow box can be traced uniquely back to a point on the local section.

Proposition 14.2.3. Suppose $\phi_{t_0}(Z_0) = X_0$ in the same context as above, except that t_0 is not required to be small. Then there exists an open set U with $Z_0 \in U$ and a function $\tau \in C^0(U)$ such that

$$\phi_{\tau(Z)}(Z) \in S, \quad \forall Z \in U,$$

and $\tau(Z_0) = t_0$.

Geometrically, starting from $Y_0 \in S$, the forward orbit $\phi_t(Y_0)$ crosses S in the direction consistent with $F(X_0)$.

relate to picture

$\tau(Z)$ is the time that the solution hits $L(X_0)$ starting at Z .

Proof. Recall that $L(X_0)$ is the line through X_0 perpendicular to $F(X_0)$, so we can write

$$L(X_0) = \{X_0 + uv_0 : u \in \mathbb{R}\}.$$

Equivalently, using $v_0 \perp F(X_0)$,

$$Y \in L(X_0) \iff (Y - X_0) \cdot F(X_0) = 0.$$

The idea is to encode the condition “the orbit hits $L(X_0)$ ” as a scalar equation, and then apply the Implicit Function Theorem to solve for the hitting time as a function of the initial condition.

Define

$$H(z, t) = (\phi_t(z) - X_0) \cdot F(X_0).$$

Then

$$H(z, t) = 0 \iff \phi_t(z) \in L(X_0).$$

By assumption $\phi_{t_0}(Z_0) = X_0$, hence

$$H(Z_0, t_0) = (\phi_{t_0}(Z_0) - X_0) \cdot F(X_0) = 0.$$

We check the nondegeneracy condition in the t -direction:

$$\begin{aligned} \partial_t H(Z_0, t_0) &= \frac{d}{dt} \left(\phi_t(Z_0) \cdot F(X_0) \right) \Big|_{t=t_0} \\ &= F(\phi_{t_0}(Z_0)) \cdot F(X_0) \\ &= F(X_0) \cdot F(X_0) \\ &= |F(X_0)|^2 \neq 0. \end{aligned}$$

Intuitively, this says the flow crosses the section transversely at X_0 , so the hitting condition changes with time.

Therefore, by the Implicit Function Theorem, there exists an open set U with $Z_0 \in U$ and a continuous function $\tau : U \rightarrow \mathbb{R}$ such that $\tau(Z_0) = t_0$ and

$$H(Z, \tau(Z)) = 0, \quad \forall Z \in U.$$

In other words, for every initial condition Z sufficiently close to Z_0 , the trajectory intersects $L(X_0)$ (indeed in the local segment S) at time $\tau(Z)$. $\square$

Definition 14.2.4 (Poincaré map). Define the map $P : S_{X_0} \rightarrow S_{X_0}$ given by $P(X)$ as the first return point to S_{X_0} starting at X .

S_{X_0} is the local section of X_0 .

If the solution is periodic, then the map is well-defined and $P(X) = X$. It can be showed with the implicit function theorem again that the solution is defined in a neighborhood of X_0 .

If $X_0 \in \omega(X_0)$, then P is well-defined.

Definition 14.2.5 (Monotone along the solution, along a line segment). Let $X_0, X_1, \dots \in \mathbb{R}^2$ be a finite or infinite sequence of distinct points on the solution curve through X_0 . We say that the sequence is monotone along the solution if $\phi_{t_n}(X_0) = X_n$ with $0 \leq t_1 < t_2 < \dots$.

Let $Y_0, Y_1, \dots$ be a finite or infinite sequence of distinct points on a line segment $I \subset \mathbb{R}^2$. The sequence is monotone along I if Y_n is between Y_{n-1} and Y_{n+1} in the natural order along I for all $n \geq 1$.

If a sequence of points is on the intersection of a solution curve and a segment I , they may be monotone along the solution curve but not monotone along the segment, or vice versa. However, if I is a local section in the plane, then this is not possible.

Proposition 14.2.6. Consider the local section S_{X_0} and suppose $Y_i = \phi_{t_i}(X) \in S_{X_0}$ for $t_1 < t_2 < \dots$. Then $Y_1, Y_2, \dots$ are monotone along S_{X_0} , i.e. Y_n is between Y_{n-1} and Y_{n+1} .

Proof. It suffices to prove that any three points Y_0, Y_1, Y_2 are monotone.

Define Σ as the trajectory between Y_0 and Y_1 , i.e.

$$\Sigma = \{\phi_t(X) : t \in [t_0, t_1]\}.$$

Now define T be the segment of S_{X_0} between Y_0 and Y_1 . We can set a region D as a domain with boundary $\Sigma \cup T$. It has an interior and exterior.

Assume that $\phi_t(x)$ leaves the closure of D at Y_1 .⁹ Then D^c is positive invariant, because any trajectory starting from D^c cannot enter D unless it crosses Σ , which violates uniqueness of the solution. Furthermore, trajectories cannot enter through T due to the direction of the flow there. More specifically,

$$\forall z \in D^c, \quad \phi_t(z) \notin \Sigma, \phi_t(z) \notin T.$$

By our assumption of $\phi_t(x)$ leaving the closure of D at Y_1 , we have that for any $\epsilon > 0$, $\phi_\epsilon(Y_1) \in D^c$. Furthermore, since we cannot reenter D , we have that

$$Y_2 = \phi_{t_2-t_1}(Y_1) \in D^c.$$

We can decompose $S = T \cup I_0 \cup I_1$ as in the diagram, where I_0 and I_1 each have one of their endpoints at Y_0 and Y_1 . Formally, we define T to be the line segment of S with endpoints Y_0 and Y_1 define

$$I_0, I_1$$

to be the two connected components of $S_{X_0} \setminus T$, labelled so that

$$Y_0 \in \overline{I_0}, \quad Y_1 \in \overline{I_1},$$

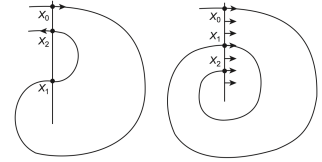


Figure 4: Two solutions crossing a straight line. On the left, X_0, X_1, X_2 is monotone along the solution but not along the line. We expect the vector field of F to yield the picture on the right when taking the line as a local section.

This is because once we have that Y_0, Y_1, Y_2 are monotone, we know that Y_2, Y_3, Y_4 must be monotone and so on.

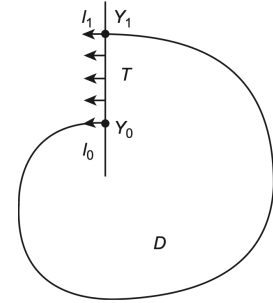


Figure 5: Solutions exit D through T .

⁹ We can also have $F(Y_1)$ entering D . This leads to D being positive invariant.

where the closures are taken inside S . into three disjoint segments.

We know $Y_2 \in S$ and $Y_2 \notin T$. We want to show that $Y_2 \in I_1$, but first have to show that $I_1 \subseteq D^c$ and $I_0 \subseteq D$. This is not trivial. If we can show this, then since $Y_2 \in I_0$ or I_1 and $Y_2 \in I_1 \subseteq D^c$, we know

$$\begin{aligned} Y_2 \in D^c \cap (I_1 \cup I_0) &= D^c \cap I_1 \\ \implies Y_2 \in I_1 &\implies \text{monotone.} \end{aligned}$$

Suppose for contradiction that I_1 contains a point of D , so there exists $p \in I_1 \cap D$. By shrinking the neighborhood of Y_1 along S if necessary, we may assume that p is arbitrarily close to Y_1 . Then

$$[\phi_\epsilon(Y_1), p] \cap \Sigma = \phi_{-\beta}(Y_1)$$

for some $\beta > 0$. This is because one end of the line segment is in D^c , while the other end is in D . Because D is open, the segment must cross the boundary; we guarantee it crossing Σ by requiring p to be close to Y_1 .

We can write

$$\begin{aligned} \phi_{-\beta}(Y_1) &= \lambda_\beta \phi_\epsilon(Y_1) + (1 - \lambda_\beta)p \\ [\phi_{-\beta}(Y_1) - Y_1] &= \lambda_\beta [\phi_\epsilon(Y_1) - Y_1] + (1 - \lambda_\beta)(p - Y_1). \end{aligned}$$

Taking the dot product of each term with $F(Y_1)$, we see that we have the left hand side is negative, the first on the right hand side is positive and the second is zero, an impossibility.

Hence $I_1 \in D^c$ and $I_2 \in I_1$. $\square$

Corollary 14.2.7. Suppose $Y \in \omega(X)$. Let Z be any point with $F(Z) \neq 0$ and let S be a local section at Z . Then the solution through Y crosses any local section S at most 1 point. Equivalently, if $\phi_{t_1}(Y) \in S$ and $\phi_{t_2}(Y) \in S$, then $\phi_{t_1}(Y) = \phi_{t_2}(Y)$.

Proof. Fix $Y \in \omega(X)$ and a local section S . Suppose for contradiction that there exist $Y_1, Y_2 \in S$ such that

$$Y_i = \phi_{t_i}(Y).$$

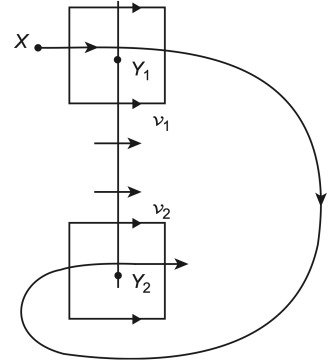
We can draw flow boxes of size σ around Y_1 and Y_2 that do not intersect, i.e. there exist $v_{1,\sigma} \cap v_{2,\sigma} = \emptyset$. The flows point in the same direction because both boxes contain part of the local section.

We know that the limiting set $\omega(X)$ is an invariant set. Hence, $Y_1, Y_2 \in \omega(X)$. Therefore we have sequences

$$\phi_{t_n}(x) \rightarrow Y_1, \quad \phi_{s_n}(x) \rightarrow Y_2,$$

where $\phi_{t_n}(x) \in v_{1,\sigma}$ and $\phi_{s_n}(x) \in v_{2,\sigma}$.

We know that the point where the segment crosses Σ will be backwards in time from Y_1 because Σ is defined as the trajectory between Y_0 and Y_1 .



We can adjust the times slightly to land on the local section:

$$\phi_{t_n+o(\sigma)}(x) \in S, \quad \phi_{s_n+o(\sigma)}(x) \in S.$$

Then

$$t_{1+o(\sigma)} < s_{k+o(\sigma)} < t_{n+o(\sigma)}$$

by definition of Y_1 and Y_2 . Label the corresponding points of $\phi_t(X)$ as A, B , and C , so by the previous proposition, we know that A, B, C are monotone along the local section. A and C are in the first box, while B is in the second box, which is disjoint from the first. This is a contradiction. $\square$

Theorem 14.2.8. (Poincaré-Bendixson) Suppose $\omega(X) \neq \emptyset$ and is topologically closed and bounded. If $\omega(X)$ does not contain an equilibrium, then $\omega(X)$ is a closed orbit.

Proof. We know that $\omega(X)$ is an invariant set. Fix $Y \in \omega(X)$, so $\phi_t(Y) \in \omega(X)$ for every $t \geq 0$, i.e. $\omega(Y) \subseteq \omega(X)$. Since $\omega(X)$ is bounded, any subsequence $\phi_{t_n}(Y)$ will have a convergent subsequence, so $\omega(Y)$ is nonempty.

Fix $Z \in \omega(Y) \subseteq \omega(X)$. Since $\omega(X)$ does not contain an equilibrium, $F(Z) \neq 0$.

We then have a local section S_Z of Z . Putting a flow box ν_σ around Z and $Z \in \omega(Y)$ implies that we have some sequence $\{t_n\}$ where $\phi_{t_n}(Y) \rightarrow Z$, we know that for some $N \in \mathbb{N}$, $n \geq N$ implies

$$\phi_{t_n}(Y) \in \nu_\sigma.$$

We can adjust the time a bit to get

$$\phi_{t_n+o(\sigma)}(Y) \in S_Z$$

for all $n \geq N$. By the previous corollary, this implies that these times must all correspond to the same point:

$$\phi_{t_N+o(\sigma)}(Y) = \phi_{t_{N+1}+o(\sigma)} = \dots$$

Putting $\tau = \min \{t_N + o(\sigma), t_{N+1} + o(\sigma), \dots\}$, we see

$$\phi_\tau(Y) = Y,$$

meaning ϕ started from Y is periodic. Let γ be the closed orbit through Y , so because $Y \in \omega(X)$, $\gamma \subseteq \omega(X)$.

We claim that $\gamma = \omega(X)$ and

$$\lim_{t \rightarrow \infty} \text{dist}(\gamma, \phi_t(x)) = 0.$$

If we have $\{\phi_t(x)\} \subseteq D$ for all $t \geq 0$, then $\omega(X) \neq \emptyset$.

should be inf but then is it still periodic?)

Fix $Y \in \omega(X)$, so $F(Y) \neq 0$. We get that there is some sequence $\tau_1 < \tau_2 < \dots$ such that

$$\phi_{t_n}(X) \rightarrow Y, \quad \phi_{t_n}(X) \in S_Y,$$

and for any $t \in (t_n, t_{n+1})$,

$$\phi_t(X) \notin S_Y.$$

Pick $0 < \epsilon < \tau$. By continuous dependence,

$$|\phi_s(Y) - \phi_{s+t_n}| \leq c_\tau |Y - \phi_{t_n}(X)|$$

for any $0 \leq s \leq 2t$ and $n \geq 1$.

Taking n large, since $\phi_\tau(Y) = Y$, we pick $s = \tau$ to get

$$|\phi_{\tau+t_n}(X) - Y| < \epsilon.$$

Furthermore,

$$|t_{n+1} - \tau - t_n| < \epsilon < \tau.$$

Note that $\phi_s(Y) \in \gamma$, so by the continuous dependence result and taking $0 \leq s \leq t_{n+1} - t_n < 2\tau$,

$$\text{dist}(\gamma, \phi_{[t_n, t_{n+1}]}(X)) \leq C_\tau |Y - \phi_{t_n}(X)|.$$

Sending $n \rightarrow \infty$ yields

$$\text{dist}(\gamma, Y) = 0.$$

□

Corollary 14.2.9. Suppose that K is a closed, bounded, and positive (negative) invariant set. Then K contains an equilibrium or a closed orbit.

Proof. Pick $X \in K$. Since K is positive invariant and closed, we have

$$\omega(X) \subseteq K, \quad \phi_t(X) \in K, \forall t \geq 0.$$

K is bounded, so any sequence t_n has a subsequence t_{n_k} where $\phi_{t_{n_k}}$ is convergence. Because K is closed, the limit is in K , so $\omega(X) \neq \emptyset$.

If $\omega(X)$ has an equilibrium, we are done. If it does not, we use the Poincaré-Bendixon theorem to see that $\omega(X)$ contains a closed orbit γ . □

Example 14.2.10. Consider

$$\frac{d}{dt}X = F(X),$$

where

$$X \cdot F(X) \big|_{\partial B_R}$$

for a positive invariant B_R . If

$$n \cdot F(X) \big|_{\partial D} < 0,$$

then D is positive invariant. ???

The first example is from the book and the second example is supposed to say that if $X \cdot F(X) > 0$ on the boundary of some region, then that region is positive invariant.

15 Textbook Notes

15.1 Teschl – 2.4: Dependence on the Initial Condition

Theorem 15.1.1. (Gronwall's inequality) Suppose $\psi(t)$ satisfies

$$\psi(t) \leq \alpha(t) + \int_0^t \beta(s)\psi(s)ds, \quad t \in [0, T]$$

for $\beta \geq 0$. Then

$$\psi(t) \leq \alpha(t) + \int_0^t \alpha(s)\beta(s) \exp\left(\int_s^t \beta(r)dr\right) ds, \quad t \in [0, T].$$

We can IBP to find also that

$$\psi(t) \leq \alpha(0) \exp\left(\int_0^t \beta(r)dr\right) + \int_0^t \alpha'(s) \exp\left(\int_s^t \beta(r)dr\right) ds.$$

Proof. Write $B(t) = \exp\left(-\int_0^t \beta(s)ds\right)$. Then

$$\begin{aligned} \frac{d}{dt} B(t) \int_0^t \beta(s)\psi(s)ds &= B(t) \cdot (-\beta(t)) \int_0^t \beta(s)\psi(s)ds + B(t)\beta(t)\psi(t) \\ &= (B(t)\beta(t)) \left(\psi(t) - \int_0^t \beta(s)\psi(s)ds \right) \\ &\leq B(t)\beta(t)\alpha(t), \end{aligned}$$

where the last line follows from the theorem assumption. Integrating (an operation that preserves the inequality, unlike differentiation), we get

$$\begin{aligned} B(t) \int_0^t \beta(s)\psi(s)ds - B(0) \int_0^0 \beta(s)\psi(s)ds &\leq \int_0^t B(s)\beta(s)\alpha(s)ds \\ \int_0^t \beta(s)\psi(s)ds &\leq \int_0^t \frac{B(s)}{B(t)} \beta(s)\alpha(s)ds \\ \psi(t) &\leq \alpha(t) + \int_0^t \beta(s)\psi(s)ds \leq \alpha(t) + \int_0^t \alpha(s)\beta(s) \frac{B(s)}{B(t)} ds \\ &= \alpha(t) + \int_0^t \alpha(s)\beta(s) \exp\left(-\int_0^s \beta(r)dr + \int_0^t \beta(r)dr\right) ds \\ &= \alpha(t) + \int_0^t \alpha(s)\beta(s) \exp\left(\int_s^t \beta(r)dr\right) ds. \end{aligned}$$

□

Theorem 15.1.2. (Continuous dependence on initial conditions and parameters) Suppose f is Lipschitz in its second argument with constant L and continuous in its first. For a given function g , if $x(t)$ and $y(t)$ are respective solutions of

f just needs to be Lipschitz on a set V in $\mathbb{R}^{1+n}$ that contains the graphs of $x(t)$ and $y(t)$

$$\begin{aligned} \frac{d}{dt}x &= f(t, x), \quad x(t_0) = x_0 \\ \frac{d}{dt}y &= g(t, y), \quad y(t_0) = y_0, \end{aligned}$$

then

$$|x(t) - y(t)| \leq |x_0 - y_0| e^{L|t-t_0|} + \frac{M}{L} (e^{L|t-t_0|} - 1),$$

where

$$M := \sup_{(t,x) \in V} |f(t, x) - g(t, x)|.$$

Proof. We have that

$$|x(t) - y(t)| \leq |x_0 - y_0| + \int_{t_0}^t |f(s, x(s)) - g(s, y(s))| ds. \quad (33)$$

We get that

$$\begin{aligned} |f(s, x(s)) - g(s, y(s))| &\leq |f(s, x(s)) - f(s, y(s))| + |f(s, y(s)) - g(s, y(s))| \\ &\leq L|x(s) - y(s)| + M. \end{aligned}$$

Plugging this back into (33), we get

$$\begin{aligned} |x(t) - y(t)| &\leq |x_0 - y_0| + \int_{t_0}^t L|x(s) - y(s)| + M ds \\ &= |x_0 - y_0| + Mt + \int_{t_0}^t L|x(s) - y(s)| ds. \end{aligned}$$

We can use Gronwall's inequality (IBP form) with

$$\alpha(t) = |x_0 - y_0| + Mt, \quad \beta(t) = L$$

to get that

$$|x(t) - y(t)| \leq |x_0 - y_0| \exp\left(\int_{t_0}^t L dr\right) + \int_{t_0}^t M \exp\left(\int_s^t L dr\right) ds,$$

so

$$\begin{aligned} |x(t) - y(t)| &\leq |x_0 - y_0| e^{Lt} + M \int_{t_0}^t e^{L(t-s)} ds \\ &= |x_0 - y_0| e^{Lt} + \frac{M}{L} (e^{L(t-t_0)} - 1). \end{aligned}$$

□

15.2 Comparison

Theorem 15.2.1. (Comparison) Suppose F is Lipschitz in its second argument and f, g are differentiable. If

$$f' \leq F(t, f(t)), \forall t \in [a, b] \quad (34)$$

$$g \text{ solves } g' = F(t, g(t)) \quad (35)$$

$$f(a) \leq g(a), \quad (36)$$

then

$$f(t) \leq g(t), \quad \forall t \in [a, b].$$

Proof. Suppose for contradiction that there exists $t_1 \in [a, b]$ such that $f(t_1) > g(t_1)$. We assume WLOG (just shrink b) that for $t > t_1$, $f(t) > g(t)$ still. Define

$$\Omega = \{t : f(t) \leq g(t)\}.$$

Ω is nonempty because it includes a , so we can put

$$t_0 := \sup\{t \in \Omega \cap [a, t_1]\}.$$

By definition, we have that for all $t \in [a, t_0]$, $f(t) \leq g(t)$, and for $t \in (t_0, t_1]$, $f(t) > g(t)$. By continuity from the left and right sides, $f(t_0) = g(t_0)$.

By assumption, F is Lipschitz in its second argument on $[a, b]$ by constant L . Note that by (34) and (35), we have that for every t ,

$$\begin{aligned} |f'(t) - g'(t)| &\leq F(t, f(t)) - F(t, g(t)) \\ f(t) - g(t) &\leq |f(t_0) - g(t_0)| + \int_{t_0}^t F(s, f(s)) - F(s, g(s)) ds \\ &\leq 0 + \int_{t_0}^t L |f(s) - g(s)| ds. \end{aligned}$$

In particular, for $t \in [t_0, t_1]$, the interval on which $f \geq g$, we have

$$f(t) - g(t) \leq \int_{t_0}^t L (f(s) - g(s)) ds.$$

We apply Gronwall's inequality (IBP form) to this with $\alpha = 0$ and $\beta = L$ to see

$$f(t) - g(t) \leq 0$$

for $t \in [t_0, t_1]$, a contradiction. □

15.3 Teschl – 2.6: Extensibility of Solutions

Lemma 15.3.1. (Gluing lemma) Suppose $f(t, x(t))$ is continuous and Lipschitz in any bounded region. Suppose $\phi_i(t)$ solves

$$\frac{d}{dt}\phi_i(t) = f(t, \phi_i(t))$$

for $t \in I_i = (a_i, b_i)$. If $t_0 \in I_1 \cap I_2$ and $\phi_1(t_0) = \phi_2(t_0)$, then the solutions agree on $I_1 \cap I_2$ and can be glued together on $I_1 \cup I_2$:

$$\phi_1(t) = \phi_2(t), \quad t \in I_1 \cap I_2,$$

$$\phi(t) = \begin{cases} \phi_1(t), & t \in I_1 \\ \phi_2(t), & t \in I_2 \end{cases}$$

is a C^1 solution to

$$\frac{d}{dt}x(t) = f(t, x(t))$$

on $I_1 \cup I_2$.

Proof. (PROOF 1) Let $I = I_1 \cap I_2 = (T_-, T_+)$ and let (t_-, t_+) be defined

$$t_- = \inf\{t \in I : \phi_1(t) = \phi_2(t)\}, \quad t_+ = \sup\{t \in I : \phi_1(t) = \phi_2(t)\}.$$

We claim that $t_+ = T_+$. Suppose towards a contradiction that $t_+ < T_+$ (this is sufficient, since $t_+ > T_+$ is not possible by definition of the maximal interval).

For every $t \in (t_-, t_+)$, $\phi_1(t) = \phi_2(t)$. ϕ_1 and ϕ_2 are defined at t_+ because $t_+ < T_+$, so $t_+ \in I$. By continuity, we see that $\phi_1(t_+) = \phi_2(t_+)$. Since F was assumed to be continuous and Lipschitz in its second argument on a bounded region¹⁰, it satisfies the conditions for the existence-uniqueness theorem. The initial conditions and differential equations for ϕ_1 and ϕ_2 coincide, so they must coincide in a neighborhood of t_+ by local existence and uniqueness, contradicting maximality of t_+ . So, $t_+ = T_+$. Similarly, $t_- = T_-$.

¹⁰ Say, a rectangle containing the graph of ϕ_1 and ϕ_2 on $[t_+ - \delta, t_+ + \delta]$

This implies that the function

$$\phi(t) = \begin{cases} \phi_1(t), & t \in I_1 \\ \phi_2(t), & t \in I_2 \end{cases}$$

is well-defined. Since it is C^1 on I_1 by looking at ϕ_1 and C^1 on I_2 by looking at ϕ_2 , it is C^1 on $I_1 \cup I_2$. It is a solution to the ODE on $I_1 \cup I_2$.

(PROOF 2) Define $I = I_1 \cap I_2$ and fix $[\alpha, \beta] \in I$ as a compact interval that contains t_0 . By F being locally Lipschitz, there exists $L \geq 0$ such that

$$\begin{aligned} \phi_1(t) - \phi_2(t) &= \int_{t_0}^t F(s, \phi_1(s)) - F(s, \phi_2(s)) ds \\ &\leq L \int_{t_0}^t |\phi_1(s) - \phi_2(s)| ds \\ &\leq L \int_{\alpha}^{\beta} |\phi_1(s) - \phi_2(s)| ds. \end{aligned}$$

Applying post-IBP Gronwall (so $\alpha(0) = 0, \alpha' = 0$),

$$\phi_1(t) - \phi_2(t) \leq 0$$

for all $t \in [\alpha, \beta]$. Since α, β were arbitrary, $\phi_1(t) = \phi_2(t)$ for all $t \in I$. We can glue and get that ϕ is C^1 from it being C^1 on I_1 and I_2 .

(PROOF 3) Define $I = I_1 \cap I_2$. Define

$$S = \{t \in I : \phi_1(t) = \phi_2(t)\}.$$

Because $\phi_1 - \phi_2$ is continuous, for any convergence sequence $\{t_n\} \rightarrow t^*$ in S ,

$$(\phi_1 - \phi_2)(t) = (\phi_1 - \phi_2)(\lim_{n \rightarrow \infty} t_n) = \lim_{n \rightarrow \infty} 0 = 0,$$

so S is closed in I .

Fixing $t \in S$, by existence-uniqueness, since $\phi_1(t) = \phi_2(t)$, they must be defined and agree on a neighborhood of t , so S is open in I .

I is a connected interval and is nonempty (it contains t_0), so $S = I$. □

Theorem 15.3.2. (*Blowup Criterion*) Let $\phi(t)$ be a solution to $\frac{d}{dt}x = f(t, x)$ on the interval (t_-, t_+) given some specific initial condition. Then there exists an extension to the interval $(t_-, t_+ + \epsilon)$ for some $\epsilon > 0$ iff there exists a sequence $\{t_n\} \in (t_-, t_+)$ such that

$$\lim_{n \rightarrow \infty} (t_n, \phi(t_n)) = (t_+, y)$$

for some $y \in \mathbb{R}$. In other words, the solution can be extended iff the graph $(t_n, \phi(t_n))$ has a finite limit point.

Proof. ($\implies$) Pick any sequence $t_n \uparrow t_+$. By continuity of ϕ , we see that there must be a limit point, which is just $(t_+, \phi(t_+))$.

($\impliedby$) Suppose we have a sequence with a finite limit point, so we have a sequence $\{t_n\} \in (t_-, t_+)$ such that $\lim_{n \rightarrow \infty} (t_n, \phi(t_n)) = (t_+, y)$.

There exists some N such that for all $n \geq N$, $|\phi(t_n) - y| \leq 1$ and $|t_n - t_+| \leq 1$. On this square

$$S := [t_+ - 1, t_+ + 1] \times [y - 1, y + 1],$$

we have that f is continuous and Lipschitz with uniform bound L . These give the bounds

$$\begin{cases} \phi'(t_n) = |f(t_n, \phi(t_n))| < M \\ |f(t, x) - f(t, z)| \leq L|x - z| \end{cases}.$$

Recall that in the existence-uniqueness proof, we required

$$h < \min\left\{\frac{1}{L}, \frac{b}{M} = \frac{1}{M}, a = 1\right\}.$$

In particular, this is not dependent on which particular t_n we want to solve the ODE in a neighborhood on, so we can fix $h > 0$ for which the solution to the ODE exists on every $B_h(t_n)$ for all $n \geq N$.

Since $t_n \uparrow t_+$, there exists t_M such that $|t_+ - t_M| < h$, so $t_M + h > t_+$. Then by the existence-uniqueness theorem, we can solve the ODE on an interval including t_+ . By the gluing lemma, we have a C^1 solution defined on $(t_-, t_+ + \epsilon)$ for some $\epsilon > 0$. □

Corollary 15.3.3. We see that the solution cannot be extended iff

$$\lim_{t_n \rightarrow \infty} f(t, \phi(t_n)) = \infty$$

for every sequence $t_n \uparrow t_+$.

By definition of $\liminf$, the solution can be extended iff

$$\left| \liminf_{t \rightarrow t_+} f(t, \phi(t)) \right| < \infty.$$

15.4 Boyce, DiPrima, Meade – 9.2 Autonomous Systems and Stability

Consider

$$\frac{d}{dt}x = f(x, y), \quad x(t_0) = x_0, \quad \frac{d}{dt}y = g(x, y), \quad y(t_0) = y_0,$$

which we write as

$$\frac{d}{dt}x = F(x), \quad x(t_0) = x_0$$

for $x : \mathbb{R} \rightarrow \mathbb{R}^2$ and $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$.

This system is called autonomous because f and g do not depend on the independent variable t but instead only on the dependent variables x and y .

Autonomous systems are interesting because they have an associated vector field that is independent of time. Consequently, there is only one trajectory passing through each point in the phase plane. All solutions that satisfy a given initial condition must lie on the same trajectory, regardless of the time at which they pass through the initial condition's point.

The following definitions are in the context of the autonomous system

$$\frac{d}{dt}x = F(x), \quad x(t_0) = x_0.$$

Definition 15.4.1 (Critical point). A point x is called a critical point of the autonomous system if $F(x) = 0$. These are also called constant or equilibrium solutions of the system of differential equations.

Definition 15.4.2 (Stable, unstable). A critical point x^* is stable if for any $\epsilon > 0$, there exists $\delta > 0$ such that every solution $x = x(t)$ that satisfies

$$\|x(t_0) - x^*\| < \delta$$

satisfies

$$\|x(t) - x^*\| < \epsilon$$

for all $t \geq t_0$.

A critical point is unstable if it is not stable.

Definition 15.4.3 (Asymptotically stable). A critical point is asymptotically stable if

Under the assumption that F is locally Lipschitz, the solution cannot blow up, so we are guaranteed existence for all $t \geq t_0$.

The precondition of stability is needed to exclude families of trajectories that have members who start arbitrarily close to x^* then depart an arbitrarily large distance before eventually tending to x^* .

- (i) it is stable, and
- (ii) there exists $\delta > 0$ such that if $\|x(t_0) - x^*\| \leq \delta$, then

$$\lim_{t \rightarrow \infty} x(t) = x^*.$$

15.5 Boyce, DiPrima, Meade – Locally Linear Systems

Consider

$$\frac{d}{dt}x = Ax$$

for a constant matrix $A \in \mathbb{R}^n \times \mathbb{R}^n$.

Theorem 15.5.1. (Linear stability of steady state) A critical point x^* of the system is

- (i) asymptotically stable if all eigenvalues of A have negative real part;
- (ii) stable and not asymptotically stable if all eigenvalues are purely imaginary; and
- (iii) unstable if at least one eigenvalue of A has positive real part.

15.6 Braun – 4.3: Stability of Equilibrium Solutions

Consider the equation

$$\frac{d}{dt}X = AX + g(X), \quad X \in \mathbb{R}^n.$$

Assume g satisfies $\lim_{x \rightarrow 0} \frac{\|g(x)\|}{\|x\|} = 0$ and the quotient is continuous, i.e. $\frac{d}{dt}X$ is C^1 . We see that since $g(0) = 0$, $X(t) = 0$ is an equilibrium solution of the ODE. We want to determine the stability of this equilibrium at the origin.

Aside 15.6.1. We recall some real analysis facts. Suppose $\frac{d}{dt}X = F(X)$ and F is differentiable. We can Taylor approximate around some point X_0 with

$$\frac{d}{dt}X = F(X_0) + \nabla F(X_0)(X - X_0) + G(X - X_0)$$

for some function G such that

$$\lim_{X \rightarrow X_0} \frac{\|G(X - X_0)\|}{\|X - X_0\|} = 0.$$

Similarly, if $F \in C^2$, then G satisfies

$$\lim_{X \rightarrow X_0} \frac{\|G(X - X_0)\|}{\|X - X_0\|^2} = 0.$$

We say that $G(X - X_0) = o(\|X - X_0\|)$ and $G(X - X_0) = o(\|X - X_0\|^2)$ respectively.

g is big- O of $\|X - X_0\|^k$ at X_0 if there exist $C \in \mathbb{R}$ and $\delta > 0$ such that

$$g(X - X_0) \leq C\|X - X_0\|^k$$

for all $\|X - X_0\| < \delta$. We see that by taking $\epsilon = 1$ in the limit definition so eventually $\frac{\|g(h)\|}{\|h\|^k} < 1$, we can multiply by $\|h\|^k$ to show

$$g(h) = o(h^k) \implies g(h) = O(h^k).$$

However, the converse is not true; consider $g(h) = |h|$.

Theorem 15.6.2. Suppose

$$\frac{d}{dt}x = Ax + g(x), x \in \mathbb{R}^n,$$

where $g \in C^2$ and A has eigenvalues $\lambda_1, \dots, \lambda_n$. Let x^* be an equilibrium, so $Ax^* + g(x^*) = 0$.

- (i) The equilibrium solution $x(t) = 0$ is asymptotically stable if $\lambda_1, \dots, \lambda_n$ all have negative real parts.
- (ii) The equilibrium solution $x(t) = 0$ is unstable if $\operatorname{Re}(\lambda_i) > 0$ for at least one i .
- (iii) The stability of $x(t) = 0$ cannot be determined if all eigenvalues have nonpositive real part but at least one eigenvalue has zero real part.

Proof. (ASYMPTOTIC STABILITY.) Suppose that $\operatorname{Re}(\lambda_1), \dots, \operatorname{Re}(\lambda_n) < 0$. By the Duhamel formula for nonlinear first order systems, we can write a solution $x(t)$ of the nonlinear ODE as

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-s)}g(x(s))ds.$$

To show asymptotic stability at the origin, we want to show that $\|x(t)\| \rightarrow 0$ as $t \rightarrow \infty$ for some $x(0)$ close to the origin.

Because A has eigenvalues all with negative real part, we can find $K, \alpha > 0$ such that

$$\|e^{At}x(0)\| \leq Ke^{-\alpha t}\|x(0)\|, \quad \|e^{A(t-s)}g(x(s))\| \leq Ke^{-\alpha(t-s)}\|g(x(s))\|.$$

Because $\frac{g(x)}{\|x\|}$ vanishes at the origin and is continuous (and hence bounded on closed disks), there exists $\sigma > 0$ such that for any $\|x\| \leq \sigma$,

$$\|g(x)\| \leq \frac{\alpha}{2K}\|x\|.$$

Plugging these bounds into the Duhamel formula, we get that

$$\begin{aligned} \|x(t)\| &\leq \|e^{At}x(0)\| + \int_0^t \|e^{A(t-s)}g(x(s))\|ds \\ &\leq Ke^{-\alpha t}\|x(0)\| + \frac{\alpha}{2} \int_0^t e^{-\alpha(t-s)}\|x(s)\|ds \\ \implies e^{\alpha t}\|x(t)\| &\leq K\|x(0)\| + \frac{\alpha}{2} \int_0^t e^{\alpha s}\|x(s)\|ds \end{aligned}$$

See exercise 4.2.17 in Braun

when $\|x(s)\| \leq \sigma$. To simplify, we write $d(t) = e^{\alpha t} \|x(t)\|$ to get

$$d(t) \leq K\|x(0)\| + \frac{\alpha}{2} \int_0^t d(s) ds.$$

We can apply Gronwall's inequality (IBP form) with $\alpha(t) = K\|x(0)\|$ and $\beta(t) = \frac{\alpha}{2} > 0$ to get

$$\begin{aligned} d(t) &\leq \alpha(0) \exp\left(\int_0^t \beta(s) ds\right) + \int_0^t \alpha'(s) \exp\left(\int_s^t \beta(r) dr\right) ds \\ &= K\|x(0)\| \exp\left(\int_0^t \frac{\alpha}{2} ds\right) + \int_0^t 0 \cdot \exp\left(\frac{\alpha}{2} dr\right) ds \\ &= K\|x(0)\| e^{\frac{\alpha}{2} t}, \end{aligned}$$

which we can substitute $d(t)$ back out to get

$$e^{\alpha t} \|x(t)\| \leq K\|x(0)\| e^{\frac{\alpha}{2} t},$$

which implies

$$\|x(t)\| \leq K\|x(0)\| e^{-\frac{\alpha}{2} t},$$

as long as $\|x(s)\| \leq \sigma$ for all $0 \leq s \leq t$. If we require $\|x(0)\| \leq \frac{\sigma}{K}$, then this is satisfied, so then we have

$$\|x(t)\| \leq K\|x(0)\| e^{-\frac{\alpha}{2} t} \leq K\|x(0)\|.$$

The second inequality gives us stability (we can contain $x(t)$ in a ball by choosing $x(0)$ to be in a small enough ball), and the first inequality shows the other condition for asymptotic stability ($x(t)$ goes to the origin in the infinite horizon).

□

15.7 Hirsch, Smale, Devaney – 9: Global Nonlinear Techniques

Consider

$$\frac{d}{dt} X = F(X), \quad X = (x_1, \dots, x_n).$$

Definition 15.7.1 (Nullcline). The x_j -nullcline is the set of points where $x'_j = f_j(x_1, \dots, x_n) = 0$.

The nullclines decompose $\mathbb{R}^n$ into a collection of open sets, in each of which the vector field points in a certain direction in relation to the standard basis vectors. These regions are called basic regions. The intersection of all nullclines yield the equilibrium points.

Definition 15.7.2 (Hyperbolic). We say that a matrix A is hyperbolic if none of its eigenvalues has real part 0.

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^1 and consider the autonomous system

$$Y' = F(Y).$$

An equilibrium point Y^* is called hyperbolic if the Jacobian matrix $J_F(Y^*)$ is hyperbolic. In this case we also say that the system is hyperbolic at Y^* .

The linearization of $Y' = F(Y)$ at the equilibrium Y^* is the linear system

$$Z' = AZ, \quad A := DF(Y^*) = J_F(Y^*),$$

and Y^* is hyperbolic iff this linear system is hyperbolic.

It's easy to determine the stability of an equilibrium point when it is hyperbolic; we can use the theorem on nonlinear stability, in which case the analysis just depends on the magnitude of the real components of the eigenvalues of the Jacobian/total derivative.

When the equilibrium point is not hyperbolic, we must use Lyapunov's method.

Definition 15.7.3 (Basin of attraction). The basin of attraction of an asymptotically stable equilibrium point is the set of all initial conditions with solutions that tend to the equilibrium point.

Definition 15.7.4 (Lyapunov function). Let $O \subseteq \mathbb{R}^n$ be open and $L : O \rightarrow \mathbb{R}$ be differentiable. Suppose $X^* \in O$, where X^* is an equilibrium of the system

$$\frac{d}{dt}X = F(X).$$

A scalar function $L : O \rightarrow \mathbb{R}$, which we call a Lyapunov candidate, has derivative along the vector field given by

$$\dot{L}(X) := \nabla L(X) \cdot F(X).$$

Note that this orbital derivative is defined as a function of the state X . However, if $z(t)$ is a particular solution to the system, we conveniently have

$$\dot{L}(z(t)) := \nabla L(z(t)) \cdot F(z(t)) = \nabla L(z(t)) \cdot \frac{d}{dt}z(t) = \frac{d}{dt}L(z(t))$$

by the chain rule; this is the reason for defining $\dot{L}$ in this way.

If L satisfies (i) and (ii) in the following theorem, then it is a Lyapunov function. If it also satisfies (iii), then it is a strict Lyapunov function.

Theorem 15.7.5. (Lyapunov stability) Let X^* be an equilibrium point for $\frac{d}{dt}X = F(X)$. Let $L : O \rightarrow \mathbb{R}$ be a differentiable function defined on an open set O that contains X^* . If

(i) $L(X^*) = 0$ and $L(X) > 0$ for $X \neq X^*$ and

(ii) $\dot{L} \leq 0$ in $O \setminus \{X^*\}$,

then X^* is stable. Furthermore, if

$$(iii) \quad \dot{L} < 0 \text{ in } O \setminus \{X^*\},$$

then X^* is asymptotically stable.

15.8 Hirsch, Smale, Devaney – 10.5: The Poincaré–Bendixson Theorem

Theorem 15.8.1. (Poincaré–Bendixson) Suppose that $\omega(X)$ is a nonempty, topologically closed, and bounded limit set of a planar system of differential equations that contains no equilibrium point. Then Ω is a closed orbit.

Proof. Let $Y \in \omega(X)$. Our first goal is to show Y is on a closed orbit. The second goal is to then show that this closed orbit is actually $\omega(X)$.

Since $Y \in \omega(X)$, we know $\omega(Y) \subseteq \omega(X)$. $\omega(Y)$ is nonempty because $\omega(X)$ is bounded. In particular, any sequence of the orbit starting from Y is contained in $\omega(X)$ and also has a convergent subsequence with limit point in $\omega(X)$ since $\omega(X)$ is compact. By definition, this limit point is in $\omega(Y)$, since we started the orbit at Y .

Fix $Z \in \omega(Y)$ and let S be a local section at Z . Let ν be a flow box associated with S . By definition of $Z \in \omega(Y)$, there exists a sequence $t_n \rightarrow \infty$ such that $\phi_{t_n}(Y) \rightarrow Z$. Hence, there are infinitely many $\phi_{t_n}(Y) \in \nu$. Picking just two of these and adding terms on the scale of $o(\sigma)$, we can find r, s such that $R > s$ and $\phi_r(Y), \phi_s(Y) \in S$.

Because we have $Y \in \omega(X)$, the trajectory through Y can only cross a local section at no more than one point. This implies that $\phi_r(Y) = \phi_s(Y)$, so $\phi_{r-s}(Y) = Y$ with $r - s > 0$. Because $\omega(X)$ contains no equilibria, Y must be on a closed (periodic) orbit. $\square$